

Supporting Information for: "Vapor–Liquid Interfacial Properties of CO₂ Mixtures for Sequestration Applications: Molecular Simulations, Classical Density Functional Theory, and Equations of State"

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The following items are presented in this Supporting Information:

1. Force field parameters of molecules used in molecular dynamics (MD) simulations (Tables S1 to S6).
2. The Perturbed Chain Statistical Associating Fluid Theory (PC-SAFT) equation of state (EoS) parameters of molecules used in Vapor-Liquid Equilibrium (VLE) and interfacial properties computations using FeOs (Table S7).
3. Empirical correlation model used to compute interfacial tensions (IFTs) of pure components (Table S8).
4. VLE of CO₂ computed from MD simulations (Tables S9 to S10).
5. Computations of IFT from MD simulations
6. VLE and interfacial properties data of pure CO₂ (Tables S11 to S16).
7. Experimental phase equilibria data of binary CO₂ mixtures (Tables S17 to S20).
8. Binary interaction parameters (k_{ij} s) of CO₂ mixtures used in the PC-SAFT EoS for VLE and interfacial properties computation (Tables S21 to S24).
9. Interfacial properties of binary CO₂ mixtures.
10. Phase equilibria and interfacial properties data of binary CO₂ mixtures (Tables S25 to S37).
11. Superheat limit (SHL) and Supercool limit (SCL) data of pure CO₂ and binary CO₂ mixtures (Tables S38 to S44).

S1 Force field parameters of molecules used in molecular dynamics (MD) simulations

Table S1: The atom types and coordinates of the three-site TraPPE CO₂ molecule (O=C=O).¹ The CO₂ bond angle used in the TraPPE model is 180° (O=Ĉ=O).

Atom type	$x / [\text{\AA}]$	$y / [\text{\AA}]$	$z / [\text{\AA}]$
O	0.00	0.00	0.00
C	1.16	0.00	0.00
O	2.32	0.00	0.00

Table S2: The TraPPE force field parameters for three-site CO₂.¹ ε and σ are the Lennard-Jones parameters and q are the partial charges.

Atom type	$\varepsilon/k_B / [\text{K}]$	$\sigma / [\text{\AA}]$	$q / [e]$
O	79.1	3.05	-0.35
C	27.0	2.80	0.70

Table S3: The atom types and coordinates of the two-site N₂ molecule (N≡N), adopted from Vrabec et al.²

Atom type	$x / [\text{\AA}]$	$y / [\text{\AA}]$	$z / [\text{\AA}]$
N	0.00	0.00	0.00
N	1.05	0.00	0.00

Table S4: The force field parameters of two-site N₂, adopted from Vrabec et al.² ε and σ are the Lennard-Jones parameters and q are the partial charges.

Atom type	$\varepsilon/k_B / [\text{K}]$	$\sigma / [\text{\AA}]$	$q / [e]$
N	34.9	3.32	0.00

Table S5: Bond-stretching and angle-bending force constants in units normalized by k_B . For CO₂, parameters are taken from Pérez-Blanco and Maginn,³ and for N₂, from Wang et al.⁴

Molecule	$k_r/k_B / [\text{K } \text{\AA}^{-2}]$	$k_\theta/k_B / [\text{K rad}^{-2}]$
CO ₂	5.18×10^5	2.8×10^4
N ₂	8.02×10^5	—

Table S6: Force field parameters of single-site Ar, H₂, and CH₄. Force fields of García-Pérez et al.⁵ and Köster et al.⁶ were adopted for Ar and H₂, respectively. The TraPPE^{1,7} force field was used to model CH₄. ε and σ are the Lennard-Jones parameters and q are the partial charges.

Atom type	ε/k_B / [K]	σ / [Å]	q / [e]
Ar	115.0	3.38	0.00
H ₂	25.80	3.04	0.00
CH ₄	148.0	3.73	0.00

S2 The Perturbed Chain Statistical Associating Fluid Theory (PC-SAFT) equation of state (EoS) parameters of molecules used in Vapor-Liquid Equilibrium (VLE) and interfacial properties computations using FeOs

Table S7: PC-SAFT EoS parameters of CO₂, N₂, Ar, H₂, and CH₄. In the PC-SAFT EoS, molecules are represented by a chain of spherical segments m interacting via a 12–6 Lennard-Jones potential. M and m are the molar weight and number of spherical segments of molecules. ε and σ are the Lennard-Jones parameters. The PC-SAFT EoS parameters were taken from the literature, as indicated in the last column.

Component	M / [g mol ⁻¹]	m / [-]	ε/k_B / [K]	σ / [Å]	References
CO ₂	43.99	2.531	153.3	2.579	Esper et al. (2023) ⁸
N ₂	28.01	1.238	89.41	3.300	Esper et al. (2023) ⁸
Ar	39.96	1.000	117.8	3.378	Esper et al. (2023) ⁸
H ₂	2.016	1.000	51.34	3.002	Eller et al. (2022) ⁹
CH ₄	16.03	1.000	150.1	3.701	Esper et al. (2023) ⁸

S3 Empirical correlation model used to compute interfacial tension (IFT) of pure components

IFTs of pure fluids were computed using Guggenheim–Katayama equation^{10–13} of the form

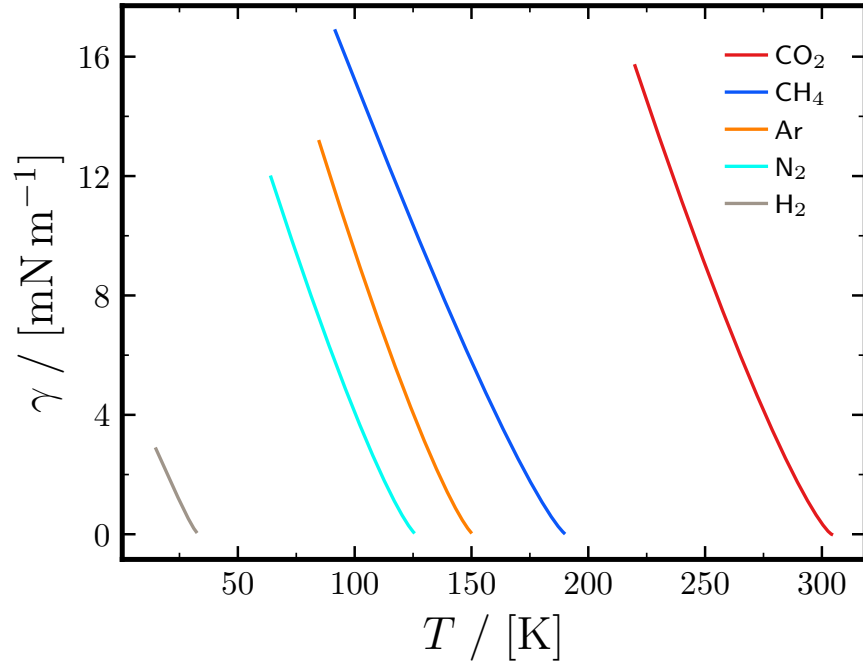
$$\gamma(T) = \sum_{i=0}^{k-1} s_i \left(1 - \frac{T}{T_c}\right)^{n_i}, \quad (\text{S1})$$

where k denotes the total number of terms in the summation, s_i and n_i are fluid-specific fit parameters, and T_c is the critical temperature of the fluid.^{14,15} The semi-empirical equation of the form shown in Eq. (S1) is the same as that used in REFPROP v10.¹⁰ For CO₂, fit parameters were adopted from Cachadiña et al. (2015)¹⁴ and for N₂, Ar, H₂, and CH₄ were adopted from Mulero et al. (2012).¹⁵ The fit parameters of CO₂, N₂, Ar, H₂, and CH₄ are listed in Table S8.

Table S8: Semi-empirical correlation fit parameters s_i and n_i for pure fluids, used in Eq. (S1) to compute IFTs. k is the number of summation terms in the correlation, and T_c is the critical temperature. Each fluid requires a single or multiple fit coefficients (s_i, n_i) pairs depending on the number of summation terms k .

Component	k / [-]	T_c / [K]	s_i / [mN m ⁻¹]	n_i / [-]	References
CO ₂	2	304.21	0.084 588	1.2814	Cachadiña et al. (2015) ¹⁴
			−0.025 372	3.9648	
N ₂	1	126.192	0.028 980	1.2460	Mulero et al. (2012) ¹⁵
Ar	1	150.687	0.037 000	1.2500	Mulero et al. (2012) ¹⁵
			−1.416 500	3.0020	
H ₂	3	33.145	0.746 383	0.6598	Mulero et al. (2012) ¹⁵
			0.675 625	0.6191	
			0.038 250	1.1910	
CH ₄	3	190.564	−0.006 024	5.4220	Mulero et al. (2012) ¹⁵
			−0.000 707	0.6161	

(a)



(b)

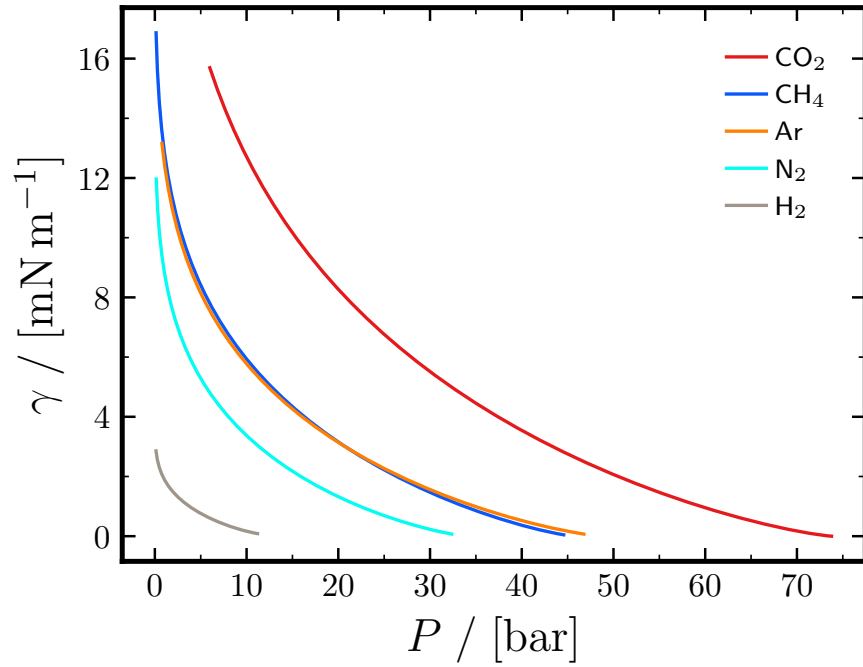


Fig. S1: The comparison of IFTs of pure CO_2 , N_2 , Ar, H_2 , and CH_4 (a) as a function of temperature and (b) as a function of saturation pressure ($P_{\text{sat}}(T)$) computed using Eq. (S1) in REFPROP v10.¹⁰

S4 VLE of CO₂ computed from MD simulations

Using the coexistence densities computed from MD simulations, the critical temperature T_c of CO₂ was calculated from the scaling law,¹⁶

$$\rho^L - \rho^V = A(T - T_c)^\beta \quad (\text{S2})$$

where ρ^L is the liquid density, ρ^V is the vapor density, and A is a fit parameter. The critical density ρ_c corresponding to T_c was calculated using the law of rectilinear diameter,¹⁷

$$\frac{\rho^L - \rho^V}{2} = \rho_c + B(T - T_c) \quad (\text{S3})$$

where ρ_c is the critical density, β is the critical exponent ($\beta = 0.32$ for three-dimensional systems¹⁸), and B is a fit parameter. Fit parameters A and B used in this work to estimate T_c and ρ_c , respectively, are listed in Table S9. The computed T_c and ρ_c of CO₂, obtained using Eq. (S2) and Eq. (S3), are presented in Table S10, in comparison with values from the Span–Wagner EoS¹⁹ and the PC-SAFT EoS.^{20,21}

To compute the critical saturation pressure P_c^{sat} corresponding to T_c of CO₂, the Antoine empirical equation²² of the form,

$$\log_{10} \left(\frac{P^{\text{sat}}}{P_0} \right) = C - \frac{D}{E + T} \quad (\text{S4})$$

was used, where P_0 is the reference pressure, taken as 1 bar. C , D , and E are empirical constants obtained by nonlinear fitting of the saturation pressure, P^{sat} , obtained as a function of T from MD simulations. Using the fit coefficients, the Eq. (S4) was evaluated at T_c to extrapolate the vapor pressure curve and estimate P_c^{sat} . Fit parameters C , D , and E used in this work to estimate P_c^{sat} are listed in Table S9 and P_c^{sat} is given in Table S10 in comparison with values from the Span–Wagner EoS Span and Wagner and the PC-SAFT EoS.^{20,21}

Table S9: Empirical fit parameters A , B , C , D , and E obtained by fitting coexistence densities and saturation pressures (220–290 K) computed from MD simulations using Eq. (S2), Eq. (S3), and Eq. (S4) to estimate T_c and P_c^{sat} of CO_2 .

T / [K]	A / [$\text{kg m}^{-3}\text{K}^{-0.32}$]	B / [$\text{kg m}^{-3}\text{K}^{-1}$]	C / [-]	D / [K]	E / [K]
220–290	471.29	−1.34	4.13	549.37	−59.67

Table S10: Computed critical temperature (T_c), critical density (ρ_c), and critical saturation pressure (P_c^{sat}) of CO_2 , obtained from fits to coexistence densities and vapor–pressure data of MD simulations in the range 220–290 K (in 10 K interval), Span–Wagner EoS Span and Wagner, and the PC-SAFT EoS.^{20,21}

Method	T_c / [K]	ρ_c / [kg m^{-3}]	P_c^{sat} / [bar]
MD	303.27	471.29	74.16
Span and Wagner EoS ¹⁹	304.13	467.60	73.77
PC-SAFT EoS ^{20,21}	309.15	454.50	82.38

S5 Computations of IFT from MD simulations

The IFT of a system can be computed from the differences between the normal and tangential components of the pressure tensor over the planar interface.²³ In MD simulations, the ensemble average of the normal and tangential components of the pressure tensor for the whole volume of the simulation box were used to compute IFT. In a parallelepiped simulation box with $L_x = L_y = 40 \text{ \AA}$ and L_z varied between 5 and 25 times L_x , containing two planar interfaces normal to the z -coordinate, the IFT is given by

$$\gamma = \frac{\langle L_z \rangle}{2} \left[\langle P_{zz} \rangle - \frac{\langle P_{xx} \rangle + \langle P_{yy} \rangle}{2} \right]. \quad (\text{S5})$$

The brackets $\langle . \rangle$ denote an ensemble average. $\langle P_{xx} \rangle$ and $\langle P_{yy} \rangle$ are the ensemble-averaged tangential components of the pressure tensor, while $\langle P_{zz} \rangle$ is the ensemble-averaged normal component. As an alternate to Eq. (S5), IFT of a system can be computed using the Irving-Kirkwood formulation:^{23,24}

$$\gamma = \frac{1}{2} \int_{-\infty}^{+\infty} \left[\langle P_{zz}(z) \rangle - \frac{\langle P_{xx}(z) \rangle + \langle P_{yy}(z) \rangle}{2} \right] dz. \quad (\text{S6})$$

where $\langle P_{xx}(z) \rangle$, $\langle P_{yy}(z) \rangle$, and $\langle P_{zz}(z) \rangle$ represent the ensemble-averaged pressure tensor profiles as a function of the z -coordinate. We computed these profiles by dividing the simulation box into 100–300 bins along the z -coordinate. The IFT was then determined by numerically integrating the differences between the normal and tangential component pressure profiles. To verify that both IFT evaluation methods yield consistent results, we calculated the IFT of pure CO_2 using Eqs. (S5) and (S6) at 220 K. Fig. S2a shows the density profile of pure CO_2 averaged for 100 ns at 220 K, and the corresponding $\Delta P = (P_N - P_T)$ profile is shown in Fig. S2b. As seen in Fig. S2b, statistical noise is present in the liquid slab of the simulation box. The anisotropy in ΔP observed in Fig. S2b was integrated to yield an IFT of 16.9 / mN m^{-1} for pure CO_2 at 220 K, as shown in Fig. S2c. Fig. S2c also shows two

plateaus, one at $\gamma = 16.9 \text{ mN m}^{-1}$ and the other at $2\gamma = 33.8 \text{ mN m}^{-1}$, due to the presence of two interfaces in the simulation box. This value matches the IFT estimated using Eq. (S5), as shown in Fig. 3a of the main text. We used Eq. (S5) throughout this study to compute the IFTs of pure CO_2 and binary CO_2 mixtures, taking advantage of the direct evaluation of IFT from the ensemble-averaged pressure tensor components for the entire simulation volume rather than computing pressure tensor profiles through spatial binning.

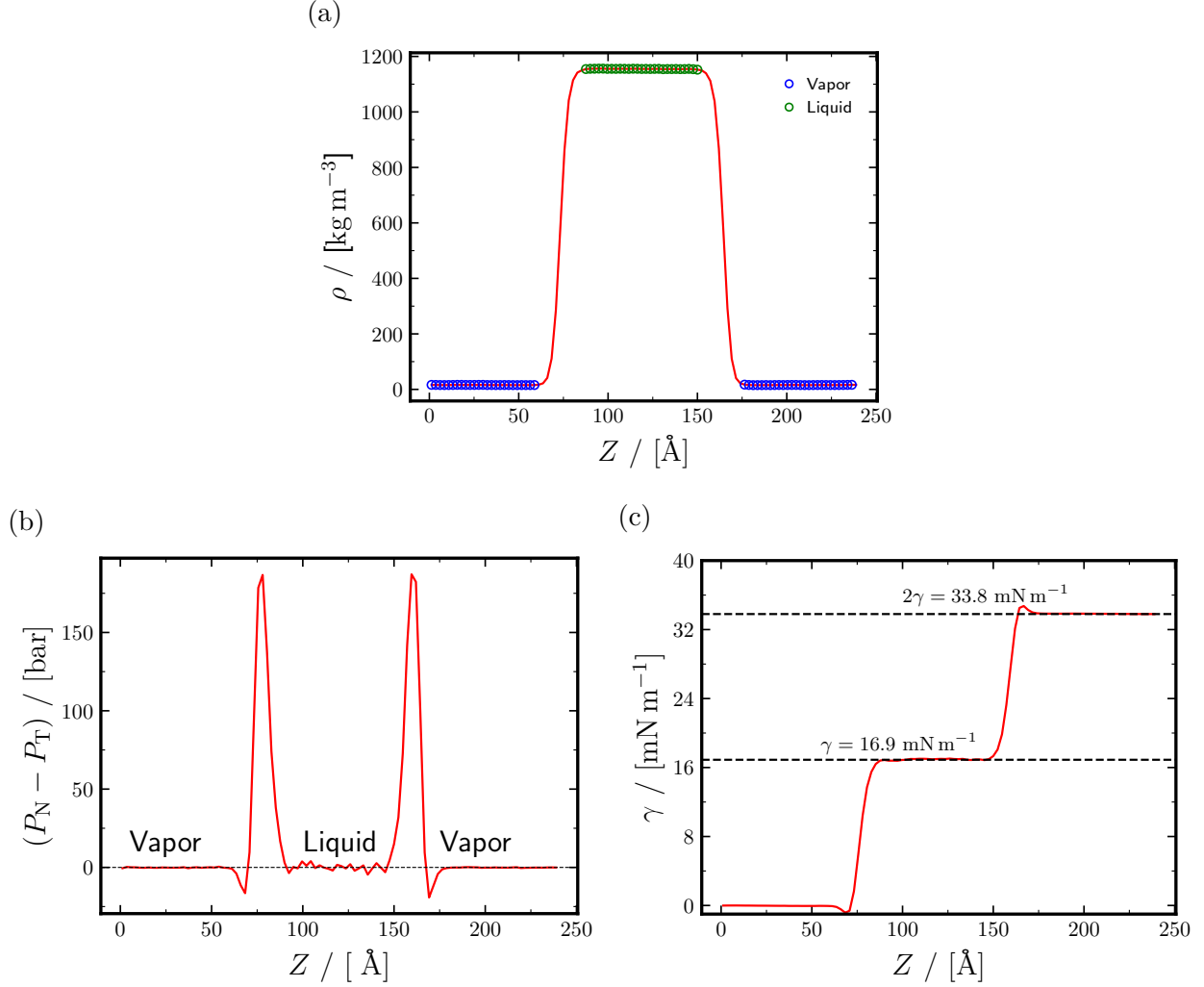


Fig. S2: (a) Density profiles of pure CO_2 obtained from MD simulations, $\rho(z)_{\text{MD}}$, at 220 K. The distinct liquid and vapor slabs in the simulation box are marked with green and blue open symbols. (b) The normal-tangential pressure difference, $\Delta P = (P_N - P_T)$, for pure CO_2 at 220 K, highlighting the pressure anisotropy that arises at the vapor-liquid interfaces. (c) Interfacial tension (IFT) evaluated as a function of the z -coordinate, obtained by integrating the ΔP profile. The appearance of two plateau regions, resulting from the two vapor-liquid interfaces in the simulation box, corresponds to an IFT value of $16.9 / \text{mN m}^{-1}$ of pure CO_2 at 220 K.

S6 VLE and interfacial properties data of pure CO₂

This section presents the raw VLE data for CO₂ obtained from MD simulations, REFPROP v10¹⁰ using the Span and Wagner EoS,^{10,19} and the PC-SAFT EoS.^{20,21} Interfacial tension (IFT) data calculated from MD simulations and from the PC-SAFT EoS combined with classical density functional theory (cDFT) are also presented. In addition, interfacial thickness values (L_{10}^{90}) obtained from both MD simulations and the PC-SAFT EoS coupled with cDFT are included.

Table S11: The Vapor–Liquid Equilibrium (VLE) data of pure CO₂ comprising saturation pressures (P^{sat}), vapor densities (ρ^V), and liquid densities (ρ^L) together with IFTs and interfacial thicknesses (L_{10}^{90}) computed from the PC-SAFT EoS^{20,21} combined with cDFT. The VLE and interfacial properties were computed at 1 K temperature increments for the temperature range 220–309 K.

$T/[\text{K}]$	$P^{\text{sat}}/[\text{bar}]$	$\rho^V/[\text{kgm}^{-3}]$	$\rho^L/[\text{kgm}^{-3}]$	$L_{10}^{90}/[\text{\AA}]$	$\gamma/[\text{mNm}^{-1}]$
220.0	6.02	15.72	1157.71	7.04	15.29
221.0	6.28	16.35	1154.29	7.10	15.07
222.0	6.54	17.00	1150.84	7.15	14.84
223.0	6.81	17.67	1147.38	7.21	14.62
224.0	7.09	18.36	1143.89	7.27	14.40
225.0	7.37	19.07	1140.38	7.33	14.17
226.0	7.67	19.81	1136.85	7.40	13.95
227.0	7.97	20.56	1133.29	7.46	13.73
228.0	8.29	21.34	1129.70	7.52	13.51
229.0	8.61	22.15	1126.10	7.59	13.29
230.0	8.94	22.97	1122.46	7.65	13.07
231.0	9.28	23.83	1118.80	7.72	12.86
232.0	9.63	24.70	1115.11	7.79	12.64
233.0	9.99	25.61	1111.39	7.86	12.42
234.0	10.36	26.53	1107.65	7.93	12.21

Table S11: The Vapor–Liquid Equilibrium (VLE) data of pure CO₂ comprising saturation pressures (P^{sat}), vapor densities (ρ^V), and liquid densities (ρ^L) together with IFTs and interfacial thicknesses (L_{10}^{90}) computed from the PC-SAFT EoS^{20,21} combined with cDFT. The VLE and interfacial properties were computed at 1 K temperature increments for the temperature range 220–309 K.

$T/[\text{K}]$	$P^{\text{sat}}/[\text{bar}]$	$\rho^V/[\text{kgm}^{-3}]$	$\rho^L/[\text{kgm}^{-3}]$	$L_{10}^{90}/[\text{\AA}]$	$\gamma/[\text{mNm}^{-1}]$
235.0	10.75	27.49	1103.87	8.00	11.99
236.0	11.14	28.47	1100.07	8.08	11.78
237.0	11.54	29.49	1096.23	8.15	11.56
238.0	11.95	30.53	1092.36	8.23	11.35
239.0	12.38	31.60	1088.46	8.31	11.14
240.0	12.81	32.70	1084.53	8.38	10.93
241.0	13.26	33.84	1080.56	8.47	10.72
242.0	13.71	35.00	1076.56	8.55	10.51
243.0	14.18	36.20	1072.52	8.63	10.30
244.0	14.67	37.44	1068.44	8.72	10.09
245.0	15.16	38.71	1064.32	8.81	9.89
246.0	15.66	40.01	1060.17	8.90	9.68
247.0	16.18	41.36	1055.97	8.99	9.48
248.0	16.71	42.74	1051.74	9.09	9.27
249.0	17.26	44.16	1047.46	9.18	9.07
250.0	17.81	45.62	1043.13	9.28	8.87
251.0	18.38	47.12	1038.76	9.38	8.67
252.0	18.97	48.67	1034.35	9.49	8.47
253.0	19.56	50.26	1029.89	9.59	8.27
254.0	20.17	51.89	1025.37	9.70	8.07
255.0	20.80	53.58	1020.81	9.81	7.87
256.0	21.44	55.31	1016.19	9.93	7.68

Table S11: The Vapor–Liquid Equilibrium (VLE) data of pure CO₂ comprising saturation pressures (P^{sat}), vapor densities (ρ^V), and liquid densities (ρ^L) together with IFTs and interfacial thicknesses (L_{10}^{90}) computed from the PC-SAFT EoS^{20,21} combined with cDFT. The VLE and interfacial properties were computed at 1 K temperature increments for the temperature range 220–309 K.

$T/[\text{K}]$	$P^{\text{sat}}/[\text{bar}]$	$\rho^V/[\text{kgm}^{-3}]$	$\rho^L/[\text{kgm}^{-3}]$	$L_{10}^{90}/[\text{\AA}]$	$\gamma/[\text{mNm}^{-1}]$
257.0	22.09	57.09	1011.52	10.05	7.48
258.0	22.76	58.92	1006.79	10.17	7.29
259.0	23.44	60.81	1002.01	10.30	7.10
260.0	24.14	62.75	997.16	10.42	6.90
261.0	24.85	64.75	992.26	10.55	6.71
262.0	25.58	66.81	987.29	10.69	6.53
263.0	26.32	68.93	982.25	10.83	6.34
264.0	27.08	71.11	977.14	10.98	6.15
265.0	27.86	73.36	971.97	11.13	5.97
266.0	28.65	75.68	966.72	11.28	5.78
267.0	29.46	78.07	961.39	11.45	5.60
268.0	30.29	80.53	955.99	11.61	5.42
269.0	31.13	83.07	950.50	11.78	5.24
270.0	31.99	85.69	944.92	11.96	5.06
271.0	32.87	88.39	939.26	12.15	4.88
272.0	33.77	91.18	933.51	12.34	4.71
273.0	34.68	94.06	927.65	12.54	4.53
274.0	35.61	97.04	921.70	12.75	4.36
275.0	36.56	100.11	915.64	12.97	4.19
276.0	37.53	103.29	909.47	13.20	4.02
277.0	38.52	106.58	903.18	13.43	3.85
278.0	39.53	109.98	896.78	13.68	3.69

Table S11: The Vapor–Liquid Equilibrium (VLE) data of pure CO₂ comprising saturation pressures (P^{sat}), vapor densities (ρ^V), and liquid densities (ρ^L) together with IFTs and interfacial thicknesses (L_{10}^{90}) computed from the PC-SAFT EoS^{20,21} combined with cDFT. The VLE and interfacial properties were computed at 1 K temperature increments for the temperature range 220–309 K.

$T/[\text{K}]$	$P^{\text{sat}}/[\text{bar}]$	$\rho^V/[\text{kgm}^{-3}]$	$\rho^L/[\text{kgm}^{-3}]$	$L_{10}^{90}/[\text{\AA}]$	$\gamma/[\text{mNm}^{-1}]$
279.0	40.56	113.50	890.24	13.94	3.52
280.0	41.60	117.15	883.57	14.22	3.36
281.0	42.67	120.94	876.76	14.51	3.20
282.0	43.76	124.86	869.80	14.81	3.04
283.0	44.87	128.94	862.68	15.13	2.89
284.0	46.00	133.17	855.40	15.47	2.73
285.0	47.15	137.58	847.94	15.83	2.58
286.0	48.32	142.16	840.28	16.22	2.43
287.0	49.52	146.94	832.43	16.62	2.29
288.0	50.74	151.92	824.37	17.06	2.14
289.0	51.98	157.12	816.07	17.53	2.00
290.0	53.24	162.57	807.52	18.03	1.86
291.0	54.53	168.27	798.71	18.58	1.72
292.0	55.84	174.25	789.61	19.17	1.59
293.0	57.17	180.54	780.19	19.81	1.46
294.0	58.53	187.17	770.43	20.51	1.33
295.0	59.91	194.16	760.29	21.29	1.21
296.0	61.32	201.57	749.73	22.15	1.09
297.0	62.75	209.43	738.70	23.12	0.97
298.0	64.21	217.81	727.14	24.37	0.86
299.0	65.70	226.78	714.99	25.44	0.75
300.0	67.21	236.42	702.15	27.02	0.64

Table S11: The Vapor–Liquid Equilibrium (VLE) data of pure CO₂ comprising saturation pressures (P^{sat}), vapor densities (ρ^V), and liquid densities (ρ^L) together with IFTs and interfacial thicknesses (L_{10}^{90}) computed from the PC-SAFT EoS^{20,21} combined with cDFT. The VLE and interfacial properties were computed at 1 K temperature increments for the temperature range 220–309 K.

$T/[\text{K}]$	$P^{\text{sat}}/[\text{bar}]$	$\rho^V/[\text{kgm}^{-3}]$	$\rho^L/[\text{kgm}^{-3}]$	$L_{10}^{90}/[\text{\AA}]$	$\gamma/[\text{mNm}^{-1}]$
301.0	68.75	246.85	688.52	28.74	0.54
302.0	70.32	258.22	673.94	30.74	0.45
303.0	71.91	270.72	658.21	33.26	0.36
304.0	73.54	284.65	641.06	36.50	0.28
305.0	75.19	300.44	622.04	40.79	0.20
306.0	76.88	318.78	600.45	46.97	0.13
307.0	78.59	341.02	574.97	57.08	0.08
309.0	82.11	423.74	485.73	61.28	0.00

Table S12: The Vapor–Liquid Equilibrium (VLE) data of pure CO₂ comprising saturation pressures (P^{sat}), vapor densities (ρ^V), and liquid densities (ρ^L) computed from the Span and Wagner EoS.¹⁹ Interfacial tension (IFTs) of pure CO₂ computed using the Guggenheim–Katayama equation^{13,15} with the coefficients reported by Cachadiña et al..¹⁴ The VLE and IFT were computed at 1 K temperature increments for the temperature range 220–304 K.

$T/[\text{K}]$	$P^{\text{sat}}/[\text{bar}]$	$\rho^V/[\text{kgm}^{-3}]$	$\rho^L/[\text{kgm}^{-3}]$	$\gamma/[\text{mNm}^{-1}]$
220.0	5.99	15.82	1166.14	15.69
221.0	6.25	16.46	1162.48	15.46
222.0	6.51	17.13	1158.81	15.23
223.0	6.78	17.82	1155.12	14.99
224.0	7.06	18.53	1151.40	14.76
225.0	7.35	19.26	1147.67	14.53
226.0	7.65	20.02	1143.92	14.30
227.0	7.95	20.79	1140.14	14.07

Table S12: The Vapor–Liquid Equilibrium (VLE) data of pure CO₂ comprising saturation pressures (P^{sat}), vapor densities (ρ^V), and liquid densities (ρ^L) computed from the Span and Wagner EoS.¹⁹ Interfacial tension (IFTs) of pure CO₂ computed using the Guggenheim–Katayama equation^{13,15} with the coefficients reported by Cachadiña et al..¹⁴ The VLE and IFT were computed at 1 K temperature increments for the temperature range 220–304 K.

$T/[\text{K}]$	$P^{\text{sat}}/[\text{bar}]$	$\rho^V/[\text{kgm}^{-3}]$	$\rho^L/[\text{kgm}^{-3}]$	$\gamma/[\text{mNm}^{-1}]$
228.0	8.27	21.59	1136.34	13.85
229.0	8.60	22.42	1132.53	13.62
230.0	8.93	23.27	1128.68	13.39
231.0	9.27	24.15	1124.82	13.16
232.0	9.63	25.05	1120.93	12.94
233.0	9.99	25.98	1117.01	12.71
234.0	10.36	26.94	1113.08	12.49
235.0	10.75	27.92	1109.11	12.27
236.0	11.14	28.93	1105.12	12.05
237.0	11.55	29.98	1101.10	11.82
238.0	11.96	31.05	1097.05	11.60
239.0	12.39	32.16	1092.98	11.38
240.0	12.82	33.30	1088.87	11.17
241.0	13.27	34.47	1084.73	10.95
242.0	13.73	35.67	1080.56	10.73
243.0	14.21	36.91	1076.36	10.51
244.0	14.69	38.18	1072.13	10.30
245.0	15.19	39.50	1067.86	10.08
246.0	15.69	40.85	1063.56	9.87
247.0	16.21	42.23	1059.22	9.66
248.0	16.75	43.66	1054.84	9.45
249.0	17.29	45.13	1050.43	9.24

Table S12: The Vapor–Liquid Equilibrium (VLE) data of pure CO₂ comprising saturation pressures (P^{sat}), vapor densities (ρ^V), and liquid densities (ρ^L) computed from the Span and Wagner EoS.¹⁹ Interfacial tension (IFTs) of pure CO₂ computed using the Guggenheim–Katayama equation^{13,15} with the coefficients reported by Cachadiña et al..¹⁴ The VLE and IFT were computed at 1 K temperature increments for the temperature range 220–304 K.

$T/[\text{K}]$	$P^{\text{sat}}/[\text{bar}]$	$\rho^V/[\text{kgm}^{-3}]$	$\rho^L/[\text{kgm}^{-3}]$	$\gamma/[\text{mNm}^{-1}]$
250.0	17.85	46.64	1045.97	9.03
251.0	18.42	48.20	1041.47	8.82
252.0	19.01	49.80	1036.93	8.61
253.0	19.61	51.45	1032.35	8.40
254.0	20.22	53.14	1027.72	8.20
255.0	20.84	54.89	1023.04	7.99
256.0	21.48	56.68	1018.32	7.79
257.0	22.14	58.53	1013.54	7.59
258.0	22.81	60.44	1008.71	7.39
259.0	23.49	62.40	1003.83	7.19
260.0	24.19	64.42	998.89	6.99
261.0	24.90	66.50	993.89	6.79
262.0	25.63	68.64	988.83	6.59
263.0	26.37	70.85	983.70	6.40
264.0	27.13	73.12	978.51	6.20
265.0	27.91	75.47	973.25	6.01
266.0	28.70	77.89	967.92	5.82
267.0	29.51	80.39	962.52	5.63
268.0	30.33	82.97	957.04	5.44
269.0	31.18	85.63	951.48	5.25
270.0	32.03	88.37	945.83	5.06
271.0	32.91	91.21	940.09	4.88

Table S12: The Vapor–Liquid Equilibrium (VLE) data of pure CO₂ comprising saturation pressures (P^{sat}), vapor densities (ρ^{V}), and liquid densities (ρ^{L}) computed from the Span and Wagner EoS.¹⁹ Interfacial tension (IFTs) of pure CO₂ computed using the Guggenheim–Katayama equation^{13,15} with the coefficients reported by Cachadiña et al..¹⁴ The VLE and IFT were computed at 1 K temperature increments for the temperature range 220–304 K.

$T/[\text{K}]$	$P^{\text{sat}}/[\text{bar}]$	$\rho^{\text{V}}/[\text{kgm}^{-3}]$	$\rho^{\text{L}}/[\text{kgm}^{-3}]$	$\gamma/[\text{mNm}^{-1}]$
272.0	33.80	94.15	934.26	4.69
273.0	34.71	97.18	928.33	4.51
274.0	35.64	100.32	922.30	4.33
275.0	36.59	103.58	916.16	4.15
276.0	37.55	106.95	909.90	3.97
277.0	38.54	110.44	903.52	3.80
278.0	39.54	114.07	897.02	3.62
279.0	40.57	117.83	890.37	3.45
280.0	41.61	121.74	883.58	3.28
281.0	42.67	125.81	876.64	3.11
282.0	43.75	130.05	869.52	2.94
283.0	44.85	134.48	862.23	2.77
284.0	45.98	139.09	854.74	2.61
285.0	47.12	143.92	847.05	2.45
286.0	48.29	148.98	839.12	2.29
287.0	49.48	154.29	830.95	2.13
288.0	50.69	159.87	822.50	1.98
289.0	51.92	165.75	813.75	1.83
290.0	53.18	171.96	804.67	1.68
291.0	54.46	178.55	795.21	1.53
292.0	55.76	185.55	785.33	1.38
293.0	57.09	193.03	774.98	1.24

Table S12: The Vapor–Liquid Equilibrium (VLE) data of pure CO₂ comprising saturation pressures (P^{sat}), vapor densities (ρ^{V}), and liquid densities (ρ^{L}) computed from the Span and Wagner EoS.¹⁹ Interfacial tension (IFTs) of pure CO₂ computed using the Guggenheim–Katayama equation^{13,15} with the coefficients reported by Cachadiña et al..¹⁴ The VLE and IFT were computed at 1 K temperature increments for the temperature range 220–304 K.

$T/[\text{K}]$	$P^{\text{sat}}/[\text{bar}]$	$\rho^{\text{V}}/[\text{kgm}^{-3}]$	$\rho^{\text{L}}/[\text{kgm}^{-3}]$	$\gamma/[\text{mNm}^{-1}]$
294.0	58.44	201.06	764.09	1.10
295.0	59.82	209.72	752.56	0.97
296.0	61.23	219.14	740.28	0.84
297.0	62.66	229.46	727.09	0.71
298.0	64.12	240.90	712.77	0.59
299.0	65.61	253.78	696.98	0.47
300.0	67.13	268.58	679.24	0.36
301.0	68.68	286.15	658.69	0.25
302.0	70.27	308.15	633.69	0.16
303.0	71.89	339.00	599.86	0.07
304.0	73.56	406.42	530.30	0.00

Table S13: The Vapor–Liquid Equilibrium (VLE) data of pure CO₂, comprising saturation pressures (P^{sat}), vapor densities (ρ^V), liquid densities (ρ^L), and their associated uncertainties computed from MD simulations, evaluated at 10 K temperature increments for the temperature range 220–309 K. At each temperature, 10 independent simulations were performed, each initialized from a different configuration. These 10 simulations were grouped into 5 blocks, from which the average values and uncertainties of the VLE properties were obtained, with the uncertainties calculated as the standard deviation of 5 blocks.

T / [K]	P^{sat} / [bar]	ρ^V / [kg m ⁻³]	ρ^L / [kg m ⁻³]
220.0	5.04 ± 0.15	16.30 ± 0.42	1151.72 ± 1.44
230.0	7.95 ± 0.21	23.77 ± 0.48	1115.86 ± 1.88
240.0	11.96 ± 0.21	33.93 ± 0.56	1076.49 ± 2.11
250.0	17.36 ± 0.24	48.15 ± 0.81	1034.04 ± 1.90
260.0	24.16 ± 0.27	66.71 ± 0.97	986.27 ± 2.02
270.0	32.44 ± 0.61	92.14 ± 1.82	933.19 ± 3.02
280.0	43.09 ± 0.58	128.10 ± 3.55	872.18 ± 2.75
290.0	54.90 ± 0.59	178.69 ± 6.02	796.67 ± 7.02

Table S14: The saturation pressures (P^{sat}), IFTs, interfacial thicknesses (L_{10}^{90}), and their associated uncertainties of pure CO_2 computed from MD simulations, evaluated at 10 K temperature increments for the temperature range 220–309 K. In this work, L_{10}^{90} of pure CO_2 were computed using two approaches: (1) using Eq. 6 of the main text and (2) by sampling the density profiles $\rho(z)_{\text{MD}}$ every 500 fs. At each temperature, 10 independent simulations were performed, each initialized from a different configuration. These 10 simulations were grouped into 5 blocks, and the average values and uncertainties of P^{sat} , IFTs, and interfacial thickness were obtained, with the uncertainties calculated as the standard deviation of 5 blocks.

$T / [\text{K}]$	$P^{\text{sat}} / [\text{bar}]$	$L_{10}^{90} / [\text{\AA}]$ (fit)	$L_{10}^{90} / [\text{\AA}]$	$\gamma / [\text{mN m}^{-1}]$
220.0	5.04 ± 0.15	5.28 ± 0.26	6.37 ± 0.06	16.91 ± 0.20
230.0	7.95 ± 0.21	5.94 ± 0.75	7.17 ± 0.09	14.45 ± 0.18
240.0	11.96 ± 0.21	7.55 ± 1.51	8.14 ± 0.10	12.09 ± 0.12
250.0	17.36 ± 0.24	7.99 ± 1.50	9.21 ± 0.10	9.75 ± 0.17
260.0	24.16 ± 0.27	9.28 ± 0.59	10.37 ± 0.10	7.38 ± 0.09
270.0	32.44 ± 0.61	11.71 ± 2.44	11.11 ± 0.07	5.39 ± 0.15
280.0	43.09 ± 0.58	12.32 ± 2.59	11.75 ± 0.16	3.45 ± 0.12
290.0	54.90 ± 0.59	20.62 ± 3.69	12.70 ± 0.14	1.90 ± 0.10

Table S15: The saturation pressures (P^{sat}), IFTs, interfacial thicknesses (L_{10}^{90}), and their associated uncertainties for pure CO₂ computed from MD simulations using the force field of Merker et al.,²⁵ evaluated at 10 K temperature increments for the range 220–309 K, taken from the Supporting Information of Stephan.²⁶

T / [K]	P^{sat} / [bar]	L_{10}^{90} / [Å]	γ / [mN m ⁻¹]
220	3.96 ± 0.15	8.1 ± 0.2	20.32 ± 0.69
230	8.64 ± 0.24	10.9 ± 0.8	15.75 ± 0.60
240	12.21 ± 0.63	11.4 ± 0.4	13.48 ± 0.47
250	18.90 ± 1.20	14.9 ± 2.5	11.30 ± 0.14
260	23.20 ± 1.60	14.3 ± 0.4	9.06 ± 0.36
270	32.00 ± 1.60	17.5 ± 1.2	7.80 ± 0.17
280	42.50 ± 0.80	21.4 ± 2.3	4.70 ± 0.11
290	50.80 ± 1.90	25.8 ± 4.8	2.77 ± 0.41

Table S16: Experimental IFT data of pure CO₂ from Quinn²⁷ and Jasper.²⁸

T / [K]	γ / [mN m ⁻¹]	Source
220.95	16.54	E. L. Quinn (1927)
230.45	13.88	E. L. Quinn (1927)
240.85	11.25	E. L. Quinn (1927)
243.15	10.08	J. J. Jasper (1972)
253.15	8.06	J. J. Jasper (1972)
263.15	6.14	J. J. Jasper (1972)
273.15	4.34	J. J. Jasper (1972)
283.15	2.67	J. J. Jasper (1972)
288.15	1.90	J. J. Jasper (1972)
293.15	1.19	J. J. Jasper (1972)
298.15	0.57	J. J. Jasper (1972)
303.15	0.07	J. J. Jasper (1972)

S7 Experimental phase equilibria data of binary CO₂ mixtures

Table S17: Experimental P_{xy} data of CO₂-CH₄ binary mixture at 230 K, 250 K, 270 K, taken from Wei et al. (1995)²⁹

T / [K]	P / [bar]	x_{CO_2} / [mol mol ⁻¹]	y_{CO_2} / [mol mol ⁻¹]	Source
230.00	10.78	0.9930	0.8447	Wei et al. (1995) ²⁹
230.00	14.53	0.9779	0.6520	Wei et al. (1995) ²⁹
230.00	17.86	0.9635	0.5450	Wei et al. (1995) ²⁹
230.00	20.92	0.9497	0.4782	Wei et al. (1995) ²⁹
230.00	25.51	0.9270	0.4071	Wei et al. (1995) ²⁹
230.00	29.16	0.9080	0.3680	Wei et al. (1995) ²⁹
230.00	35.03	0.8731	0.3230	Wei et al. (1995) ²⁹
230.00	39.68	0.8423	0.2995	Wei et al. (1995) ²⁹
230.00	45.65	0.7958	0.2763	Wei et al. (1995) ²⁹
230.00	51.16	0.7440	0.2632	Wei et al. (1995) ²⁹
230.00	56.40	0.6829	0.2551	Wei et al. (1995) ²⁹
230.00	58.85	0.6584	0.2534	Wei et al. (1995) ²⁹
230.00	60.86	0.6195	0.2539	Wei et al. (1995) ²⁹
230.00	63.53	0.5813	0.2552	Wei et al. (1995) ²⁹
230.00	65.86	0.5477	0.2589	Wei et al. (1995) ²⁹
230.00	66.63	0.5238	0.2631	Wei et al. (1995) ²⁹
230.00	67.36	0.5048	0.2662	Wei et al. (1995) ²⁹
230.00	69.01	0.4593	0.2753	Wei et al. (1995) ²⁹
230.00	69.91	0.4186	0.2996	Wei et al. (1995) ²⁹
250.00	19.67	0.9935	0.9218	Wei et al. (1995) ²⁹

Table S17: Experimental P_{xy} data of CO₂-CH₄ binary mixture at 230 K, 250 K, 270 K, taken from Wei et al. (1995)²⁹

T / [K]	P / [bar]	x_{CO_2} / [mol mol ⁻¹]	y_{CO_2} / [mol mol ⁻¹]	Source
250.00	20.71	0.9894	0.8826	Wei et al. (1995) ²⁹
250.00	23.15	0.9799	0.8048	Wei et al. (1995) ²⁹
250.00	24.88	0.9731	0.7592	Wei et al. (1995) ²⁹
250.00	27.59	0.9621	0.6999	Wei et al. (1995) ²⁹
250.00	29.64	0.9535	0.6619	Wei et al. (1995) ²⁹
250.00	31.31	0.9464	0.6345	Wei et al. (1995) ²⁹
250.00	34.46	0.9324	0.5910	Wei et al. (1995) ²⁹
250.00	38.36	0.9137	0.5467	Wei et al. (1995) ²⁹
250.00	47.92	0.8632	0.4738	Wei et al. (1995) ²⁹
250.00	53.78	0.8285	0.4441	Wei et al. (1995) ²⁹
250.00	61.50	0.7754	0.4188	Wei et al. (1995) ²⁹
250.00	67.15	0.7289	0.4086	Wei et al. (1995) ²⁹
250.00	73.99	0.6634	0.4144	Wei et al. (1995) ²⁹
250.00	76.84	0.6282	0.4239	Wei et al. (1995) ²⁹
250.00	79.83	0.5837	0.4477	Wei et al. (1995) ²⁹
250.00	80.53	0.5748	0.4592	Wei et al. (1995) ²⁹
270.00	35.58	0.9849	0.9234	Wei et al. (1995) ²⁹
270.00	36.73	0.9802	0.8953	Wei et al. (1995) ²⁹
270.00	38.83	0.9695	0.8539	Wei et al. (1995) ²⁹
270.00	41.46	0.9623	0.8222	Wei et al. (1995) ²⁹
270.00	45.06	0.9496	0.7797	Wei et al. (1995) ²⁹
270.00	49.15	0.9276	0.7339	Wei et al. (1995) ²⁹
270.00	58.51	0.8846	0.6628	Wei et al. (1995) ²⁹

Table S17: Experimental P_{xy} data of CO₂-CH₄ binary mixture at 230 K, 250 K, 270 K, taken from Wei et al. (1995)²⁹

T / [K]	P / [bar]	x_{CO_2} / [mol mol ⁻¹]	y_{CO_2} / [mol mol ⁻¹]	Source
270.00	70.93	0.8166	0.6058	Wei et al. (1995) ²⁹
270.00	78.68	0.7593	0.5939	Wei et al. (1995) ²⁹
270.00	81.97	0.7304	0.6048	Wei et al. (1995) ²⁹
270.00	82.82	0.7197	0.6113	Wei et al. (1995) ²⁹

Table S18: Experimental P_{xy} data of CO₂-Ar binary mixture at 223.07 K, 253.28 K, and 273.26 K taken from Coquelet et al. (2008)³⁰ and Lasala et al. (2016).³¹

T / [K]	P / [bar]	x_{CO_2} / [mol mol ⁻¹]	y_{CO_2} / [mol mol ⁻¹]	Source
223.07	18.95	0.9803	0.4315	Lasala et al. (2016) ³¹
223.07	28.64	0.9634	0.3122	Lasala et al. (2016) ³¹
223.07	39.90	0.9424	0.2500	Lasala et al. (2016) ³¹
223.07	48.72	0.9201	0.2256	Lasala et al. (2016) ³¹
223.07	58.57	0.8950	0.2070	Lasala et al. (2016) ³¹
223.07	68.87	0.8710	0.1952	Lasala et al. (2016) ³¹
223.07	79.20	0.8405	0.1911	Lasala et al. (2016) ³¹
223.07	93.18	0.8008	0.1930	Lasala et al. (2016) ³¹
223.07	103.96	0.7653	0.1996	Lasala et al. (2016) ³¹
223.07	116.54	0.7184	0.2140	Lasala et al. (2016) ³¹
223.07	129.59	0.6604	0.2435	Lasala et al. (2016) ³¹
223.07	140.63	0.5914	0.3013	Lasala et al. (2016) ³¹
223.07	145.52	0.5443	0.3426	Lasala et al. (2016) ³¹
253.28	25.50	0.9885	0.8235	Coquelet et al. (2008) ³⁰

Table S18: Experimental P_{xy} data of CO₂-Ar binary mixture at 223.07 K, 253.28 K, and 273.26 K taken from Coquelet et al. (2008)³⁰ and Lasala et al. (2016).³¹

T / [K]	P / [bar]	x_{CO_2} / [mol mol ⁻¹]	y_{CO_2} / [mol mol ⁻¹]	Source
253.28	37.00	0.9650	0.6270	Coquelet et al. (2008) ³⁰
253.28	49.05	0.9388	0.5224	Coquelet et al. (2008) ³⁰
253.28	61.38	0.9099	0.4584	Coquelet et al. (2008) ³⁰
253.28	73.02	0.8805	0.4232	Coquelet et al. (2008) ³⁰
253.28	83.06	0.8540	0.4062	Coquelet et al. (2008) ³⁰
253.28	91.92	0.8277	0.3985	Coquelet et al. (2008) ³⁰
253.28	100.02	0.8014	0.3974	Coquelet et al. (2008) ³⁰
253.28	107.07	0.7766	0.4017	Coquelet et al. (2008) ³⁰
253.28	112.01	0.7564	0.4061	Coquelet et al. (2008) ³⁰
253.28	117.17	0.7335	0.4121	Coquelet et al. (2008) ³⁰
253.28	122.12	0.7088	0.4237	Coquelet et al. (2008) ³⁰
253.28	124.03	0.6977	0.4280	Coquelet et al. (2008) ³⁰
253.28	126.88	0.6799	0.4398	Coquelet et al. (2008) ³⁰
253.28	129.87	0.6567	0.4554	Coquelet et al. (2008) ³⁰
273.26	38.23	0.9931	0.9420	Coquelet et al. (2008) ³⁰
273.26	47.98	0.9718	0.8144	Coquelet et al. (2008) ³⁰
273.26	58.04	0.9487	0.7363	Coquelet et al. (2008) ³⁰
273.26	68.10	0.9239	0.6750	Coquelet et al. (2008) ³⁰
273.26	78.04	0.8970	0.6423	Coquelet et al. (2008) ³⁰
273.26	87.97	0.8676	0.6222	Coquelet et al. (2008) ³⁰
273.26	97.88	0.8330	0.6066	Coquelet et al. (2008) ³⁰
273.26	104.88	0.8046	0.6078	Coquelet et al. (2008) ³⁰
273.26	110.48	0.7768	0.6193	Coquelet et al. (2008) ³⁰

Table S18: Experimental P_{xy} data of CO₂-Ar binary mixture at 223.07 K, 253.28 K, and 273.26 K taken from Coquelet et al. (2008)³⁰ and Lasala et al. (2016).³¹

T / [K]	P / [bar]	x_{CO_2} / [mol mol ⁻¹]	y_{CO_2} / [mol mol ⁻¹]	Source
273.26	114.92	0.7391	0.6471	Coquelet et al. (2008) ³⁰
273.26	115.87	0.7238	0.6658	Coquelet et al. (2008) ³⁰

Table S19: Experimental P_{xy} data of CO₂-N₂ binary mixture at 220 K, 250 K, and 273.12 K taken from Al-Sahhaf et al. (1983),³² Brown et al. (1989),³³ and Lasala et al. (2016),³¹ respectively.

T / [K]	P / [bar]	x_{CO_2} / [mol mol ⁻¹]	y_{CO_2} / [mol mol ⁻¹]	Source
220.00	15.09	0.9868	0.4482	Al-Sahhaf et al. (1983) ³²
220.00	30.75	0.9638	0.2577	Al-Sahhaf et al. (1983) ³²
220.00	45.95	0.9399	0.2026	Al-Sahhaf et al. (1983) ³²
220.00	63.13	0.9117	0.1786	Al-Sahhaf et al. (1983) ³²
220.00	82.58	0.8791	0.1719	Al-Sahhaf et al. (1983) ³²
220.00	111.14	0.8258	0.1835	Al-Sahhaf et al. (1983) ³²
220.00	124.32	0.7984	0.1972	Al-Sahhaf et al. (1983) ³²
220.00	137.96	0.7676	0.2131	Al-Sahhaf et al. (1983) ³²
220.00	155.06	0.7229	0.2484	Al-Sahhaf et al. (1983) ³²
220.00	167.06	0.6845	0.2785	Al-Sahhaf et al. (1983) ³²
250.00	25.99	0.9864	0.7439	Brown et al. (1989) ³³
250.00	36.88	0.9669	0.5761	Brown et al. (1989) ³³
250.00	45.89	0.9512	0.4999	Brown et al. (1989) ³³
250.00	55.41	0.9336	0.4506	Brown et al. (1989) ³³
250.00	65.97	0.9130	0.4151	Brown et al. (1989) ³³
250.00	75.32	0.8931	0.3951	Brown et al. (1989) ³³

Table S19: Experimental P_{xy} data of CO₂-N₂ binary mixture at 220 K, 250 K, and 273.12 K taken from Al-Sahhaf et al. (1983),³² Brown et al. (1989),³³ and Lasala et al. (2016),³¹ respectively.

T / [K]	P / [bar]	x_{CO_2} / [mol mol ⁻¹]	y_{CO_2} / [mol mol ⁻¹]	Source
250.00	96.32	0.8468	0.3777	Brown et al. (1989) ³³
250.00	111.51	0.8069	0.3837	Brown et al. (1989) ³³
250.00	119.36	0.7827	0.3937	Brown et al. (1989) ³³
250.00	134.16	0.7292	0.4273	Brown et al. (1989) ³³
250.00	140.66	0.6992	0.4563	Brown et al. (1989) ³³
273.12	45.75	0.9754	0.8355	Lasala et al. (2016) ³¹
273.12	50.68	0.9639	0.7861	Lasala et al. (2016) ³¹
273.12	57.38	0.9481	0.7322	Lasala et al. (2016) ³¹
273.12	65.03	0.9302	0.6859	Lasala et al. (2016) ³¹
273.12	74.31	0.9044	0.6466	Lasala et al. (2016) ³¹
273.12	84.44	0.8754	0.6201	Lasala et al. (2016) ³¹
273.12	93.87	0.8435	0.6095	Lasala et al. (2016) ³¹
273.12	103.94	0.8039	0.6086	Lasala et al. (2016) ³¹
273.12	109.50	0.7787	0.6204	Lasala et al. (2016) ³¹
273.12	112.06	0.7572	0.6302	Lasala et al. (2016) ³¹
273.12	113.65	0.7437	0.6391	Lasala et al. (2016) ³¹
273.12	114.85	0.7271	0.6550	Lasala et al. (2016) ³¹

Table S20: Experimental P_{xy} data of CO₂-H₂ binary mixture at 220 K, 250 K, and 273.12 K taken from Tsang and Street (1981)³⁴ and Lasala et al. (2016).³¹

T / [K]	P / [bar]	x_{CO_2} / [mol mol ⁻¹]	y_{CO_2} / [mol mol ⁻¹]	Source
220.00	9.30	0.9985	0.7102	Tsang and Street (1981) ³⁴

Table S20: Experimental P_{xy} data of CO₂-H₂ binary mixture at 220 K, 250 K, and 273.12 K taken from Tsang and Street (1981)³⁴ and Lasala et al. (2016).³¹

T / [K]	P / [bar]	x_{CO_2} / [mol mol ⁻¹]	y_{CO_2} / [mol mol ⁻¹]	Source
220.00	10.70	0.9972	0.6129	Tsang and Street (1981) ³⁴
220.00	12.70	0.9968	0.5358	Tsang and Street (1981) ³⁴
220.00	13.50	0.9965	0.4975	Tsang and Street (1981) ³⁴
220.00	22.30	0.9933	0.3245	Tsang and Street (1981) ³⁴
220.00	35.10	0.9888	0.2190	Tsang and Street (1981) ³⁴
220.00	51.30	0.9841	0.1631	Tsang and Street (1981) ³⁴
220.00	71.30	0.9782	0.1263	Tsang and Street (1981) ³⁴
220.00	110.10	0.9703	0.0945	Tsang and Street (1981) ³⁴
220.00	140.00	0.9639	0.0838	Tsang and Street (1981) ³⁴
220.00	154.60	0.9606	0.0811	Tsang and Street (1981) ³⁴
220.00	169.30	0.9573	0.0787	Tsang and Street (1981) ³⁴
220.00	187.90	0.9537	0.0791	Tsang and Street (1981) ³⁴
220.00	209.90	0.9486	0.0740	Tsang and Street (1981) ³⁴
220.00	243.40	0.9407	0.0701	Tsang and Street (1981) ³⁴
220.00	264.40	0.9353	0.0693	Tsang and Street (1981) ³⁴
220.00	275.90	0.9337	0.0697	Tsang and Street (1981) ³⁴
220.00	315.00	0.9279	0.0693	Tsang and Street (1981) ³⁴
220.00	354.00	0.9162	0.0685	Tsang and Street (1981) ³⁴
250.00	23.10	0.9956	0.8411	Tsang and Street (1981) ³⁴
250.00	30.40	0.9914	0.6804	Tsang and Street (1981) ³⁴
250.00	36.30	0.9876	0.5910	Tsang and Street (1981) ³⁴
250.00	52.00	0.9814	0.4499	Tsang and Street (1981) ³⁴
250.00	69.00	0.9738	0.3629	Tsang and Street (1981) ³⁴

Table S20: Experimental P_{xy} data of CO₂-H₂ binary mixture at 220 K, 250 K, and 273.12 K taken from Tsang and Street (1981)³⁴ and Lasala et al. (2016).³¹

$T / [\text{K}]$	$P / [\text{bar}]$	$x_{\text{CO}_2} / [\text{mol mol}^{-1}]$	$y_{\text{CO}_2} / [\text{mol mol}^{-1}]$	Source
250.00	102.90	0.9604	0.2821	Tsang and Street (1981) ³⁴
250.00	138.50	0.9455	0.2415	Tsang and Street (1981) ³⁴
250.00	172.80	0.9305	0.2178	Tsang and Street (1981) ³⁴
250.00	254.40	0.8942	0.1956	Tsang and Street (1981) ³⁴
250.00	315.40	0.8668	0.1895	Tsang and Street (1981) ³⁴
250.00	353.40	0.8512	0.1884	Tsang and Street (1981) ³⁴
250.00	496.60	0.7879	0.1940	Tsang and Street (1981) ³⁴
250.00	628.10	0.7276	0.2166	Tsang and Street (1981) ³⁴
250.00	758.80	0.6588	0.2459	Tsang and Street (1981) ³⁴
250.00	832.20	0.6113	0.2700	Tsang and Street (1981) ³⁴
250.00	865.40	0.5853	0.2874	Tsang and Street (1981) ³⁴
250.00	883.90	0.5683	0.2970	Tsang and Street (1981) ³⁴
273.10	40.83	0.9959	0.9058	Fandiño et al. (2015) ³⁵
273.10	43.65	0.9940	0.8684	Fandiño et al. (2015) ³⁵
273.10	46.73	0.9918	0.8321	Fandiño et al. (2015) ³⁵
273.10	53.01	0.9876	0.7701	Fandiño et al. (2015) ³⁵
273.10	62.54	0.9811	0.6957	Fandiño et al. (2015) ³⁵
273.10	76.13	0.9719	0.6190	Fandiño et al. (2015) ³⁵
273.10	101.33	0.9546	0.5279	Fandiño et al. (2015) ³⁵
273.10	124.58	0.9376	0.4758	Fandiño et al. (2015) ³⁵
273.10	148.67	0.9138	0.4398	Fandiño et al. (2015) ³⁵

S8 Binary interaction parameters (k_{ij} s) of CO₂ mixtures used in the PC-SAFT EoS for VLE and interfacial properties computation

The binary interaction parameters k_{ij} s were estimated using the PC-SAFT EoS^{20,21} from FeOs³⁶ by performing three separate nonlinear least-squares optimizations that minimize the deviation between experimental and computed VLE data. For each trial value of k_{ij} , the PC-SAFT EoS computes the phase equilibrium properties at the experimental temperatures, including compositions, saturation pressure, and chemical potentials. Depending on the formulation of the objective function, the residuals were defined either as the distance between computed and experimental points in the Pxy phase diagram, as the difference in chemical potentials of the coexisting phases, or as the difference between computed and experimental saturation pressures. In all cases, the optimization cost (objective) function was expressed as

$$\text{Cost}(k_{ij}) = \sum_{n=1}^N r_n(k_{ij})^2, \quad (\text{S7})$$

where $r_n(k_{ij})$ denotes the appropriate residual for the n^{th} data point. The minimization was carried out using the `scipy.optimize.least_squares` algorithm, which iteratively adjusts k_{ij} to reduce $\text{Cost}(k_{ij})$. After each optimization, the quality of the resulting k_{ij} was assessed by calculating the Mean Absolute Percentage Deviation (MAPD) for the residuals of saturation pressure, distance, and chemical potential. The computed MAPD values are reported in Tables S21 to S23 for the CO₂-CH₄, CO₂-Ar, CO₂-N₂, and CO₂-H₂ at different temperatures. The Pxy diagrams of all the binary mixtures at different temperatures are presented in Figs. S3 to S14 using the k_{ij} values listed in Tables S21 to S23. The k_{ij} values that produced the lowest MAPD relative to the experimental data were chosen to compute the phase equilibria and interfacial properties with PC-SAFT EoS + cDFT for each binary mixture. When comparing the k_{ij} values obtained from optimizing cost functions based

on distance, chemical potential, and pressure, the distance-based optimization consistently yielded the lowest MAPD. In the case of the CO₂-H₂ mixture at 220 K, the phase equilibria computations (Fig. S12) using the k_{ij} values optimized with the chemical potential and saturation pressure cost functions did not converge to the critical point. Hence, the k_{ij} values listed in Table S21 were used to compute the phase equilibria and interfacial properties. To compute the interfacial properties of the CO₂-CH₄ mixture at 220.0 K, the CO₂-Ar mixture at 220.0, 250.0, and 270.0 K, the CO₂-N₂ mixture at 270.0 K, and the CO₂-H₂ mixture at 270.0 K, the k_{ij} values listed in Table S21 were fitted using a third-order polynomial. The resulting polynomial fit, based on the k_{ij} values obtained from optimizing the distance-based cost function with experimental data,^{29–35} is shown in Fig. S15. The corresponding k_{ij} values for CO₂-CH₄ at 220.0 K, CO₂-Ar at 220.0, 250.0, and 270.0 K, CO₂-N₂ at 270.0 K, and CO₂-H₂ at 270.0 K are listed in Table S24.

Table S21: Binary interaction parameters (k_{ij} s) of CO₂-CH₄, CO₂-Ar, CO₂-N₂, and CO₂-H₂ at different temperatures obtained from optimizing the cost function based on distance. The experimental data reported in the literature^{29–35} were used for optimizing k_{ij} . Experimental data for the CO₂-CH₄ mixtures were taken from Wei et al.²⁹ at 230, 250, and 270 K. For the CO₂-Ar mixtures, the experimental data were taken from Lasala et al.³¹ at 223.07 K and from Coquelet et al.³⁰ at 253.28 and 273.26 K. Experimental data for the CO₂-N₂ mixtures at 220, 250, and 273.12 K were obtained from Brown et al.,³³ Al-Sahhaf et al.,³² and Lasala et al.,³¹ respectively. Experimental data for the CO₂-H₂ mixtures were taken from Tsang and Street³⁴ at 220 and 250 K and from Fandiño et al.³⁵ at 273.1 K.

Binary mixture	T / [K]	k_{ij} / [-]	MAPD / [%]
CO ₂ -CH ₄	230.0	0.0519	1.1850
	250.0	0.0541	1.0211
	270.0	0.0566	0.7362
CO ₂ -Ar	223.1	0.0317	1.0836
	253.3	0.0295	1.5178
	273.3	0.0240	1.6017
CO ₂ -N ₂	220.0	−0.0164	0.6170
	250.0	−0.0294	0.5076
	273.1	−0.0699	1.0277
CO ₂ -H ₂	220.0	−0.0594	3.2917
	250.0	0.0474	0.8745
	273.1	0.0538	0.3000

Table S22: Binary interaction parameters (k_{ij} s) of CO₂-CH₄, CO₂-Ar, CO₂-N₂, and CO₂-H₂ at different temperatures obtained from optimizing the cost function based on chemical potential. The experimental data reported in the literature^{29–35} were used for optimizing k_{ij} . Experimental data for the CO₂-CH₄ mixtures were taken from Wei et al.²⁹ at 230, 250, and 270 K. For the CO₂-Ar mixtures, the experimental data were taken from Lasala et al.³¹ at 223.07 K and from Coquelet et al.³⁰ at 253.28 and 273.26 K. Experimental data for the CO₂-N₂ mixtures at 220, 250, and 273.12 K were obtained from Brown et al.,³³ Al-Sahhaf et al.,³² and Lasala et al.,³¹ respectively. Experimental data for the CO₂-H₂ mixtures were taken from Tsang and Street³⁴ at 220 and 250 K and from Fandiño et al.³⁵ at 273.1 K.

Binary mixture	T / [K]	k_{ij} / [-]	MAPD / [%]
CO ₂ -CH ₄	230.0	0.0703	2.1995
	250.0	0.0682	1.8135
	270.0	0.0510	1.4662
CO ₂ -Ar	223.1	0.0586	3.3301
	253.3	0.0439	1.7721
	273.3	0.0366	1.4869
CO ₂ -N ₂	220.0	−0.0042	1.4153
	250.0	−0.0118	1.2043
	273.1	−0.0525	1.0812
CO ₂ -H ₂	220.0	0.0756	5.8186
	250.0	0.0463	3.7433
	273.1	0.0765	0.6865

Table S23: Binary interaction parameters (k_{ij} s) of CO₂-CH₄, CO₂-Ar, CO₂-N₂, and CO₂-H₂ at different temperatures obtained from optimizing the cost function based on saturation pressure. The experimental data reported in the literature^{29–35} were used for optimizing k_{ij} . Experimental data for the CO₂-CH₄ mixtures were taken from Wei et al.²⁹ at 230, 250, and 270 K. For the CO₂-Ar mixtures, the experimental data were taken from Lasala et al.³¹ at 223.07 K and from Coquelet et al.³⁰ at 253.28 and 273.26 K. Experimental data for the CO₂-N₂ mixtures at 220, 250, and 273.12 K were obtained from Brown et al.,³³ Al-Sahhaf et al.,³² and Lasala et al.,³¹ respectively. Experimental data for the CO₂-H₂ mixtures were taken from Tsang and Street³⁴ at 220 and 250 K and from Fandiño et al.³⁵ at 273.1 K.

Binary mixture	T / [K]	k_{ij} / [-]	MAPD / [%]
CO ₂ -CH ₄	230.0	0.0608	5.5625
	250.0	0.0594	3.7421
	270.0	0.0514	1.6356
CO ₂ -Ar	223.1	0.0454	11.6866
	253.3	0.0384	4.5741
	273.3	0.0336	3.4173
CO ₂ -N ₂	220.0	−0.0099	6.1761
	250.0	−0.0184	4.1366
	273.1	−0.0585	3.1974
CO ₂ -H ₂	220.0	0.0908	14.2086
	250.0	0.0495	9.7470
	273.1	0.0803	1.9750

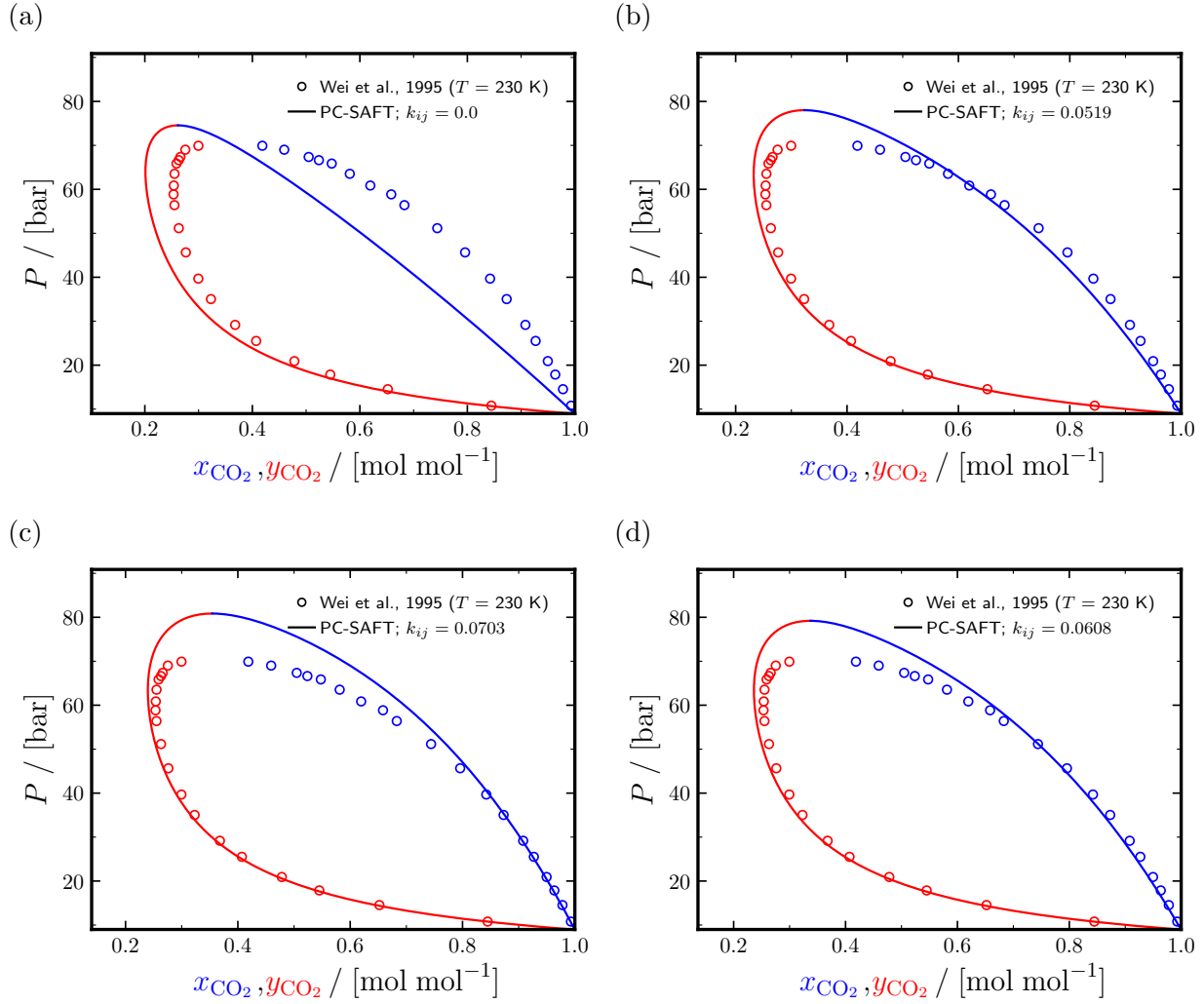


Fig. S3: Pxy diagrams of $\text{CO}_2\text{-CH}_4$ mixtures computed using the PC-SAFT EoS^{20,21} at 230 K, in comparison with the experimental data of Wei et al. (1995).²⁹ The diagrams were generated using (a) $k_{ij} = 0$, (b) $k_{ij} = 0.0519$, (c) $k_{ij} = 0.0703$, and (d) $k_{ij} = 0.0608$, where the values in (b)–(d) were obtained from cost-function optimizations performed using the experimental measurements²⁹ with distance, chemical potential, and saturation pressure criteria, respectively.

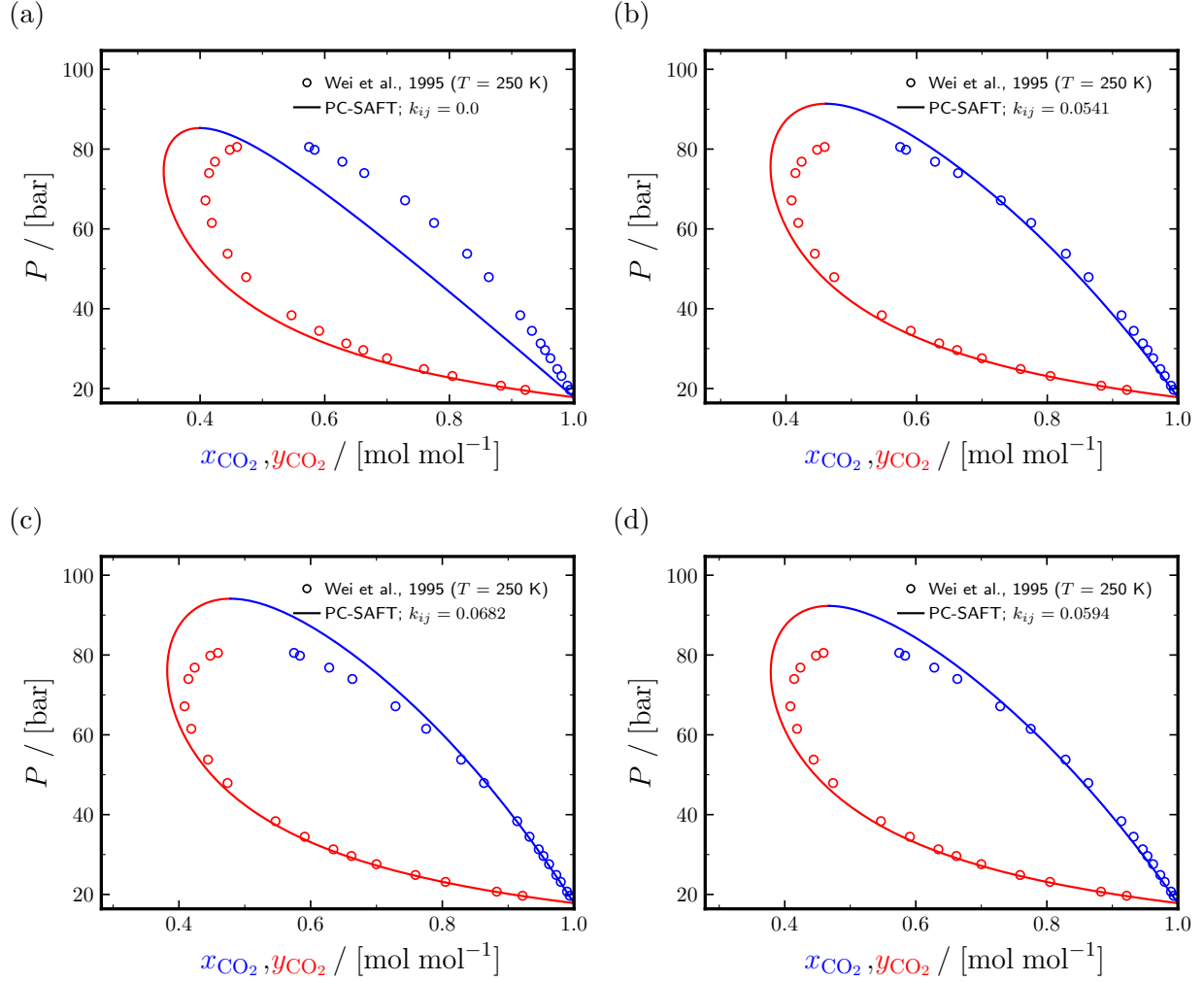


Fig. S4: Pxy diagrams of CO_2 - CH_4 mixtures computed using the PC-SAFT EoS^{20,21} at 250 K, in comparison with the experimental data of Wei et al. (1995).²⁹ The diagrams were generated using (a) $k_{ij} = 0$, (b) $k_{ij} = 0.0541$, (c) $k_{ij} = 0.0682$, and (d) $k_{ij} = 0.0594$, where the values in (b)–(d) were obtained from cost-function optimizations performed using the experimental measurements²⁹ with distance, chemical potential, and saturation pressure criteria, respectively.

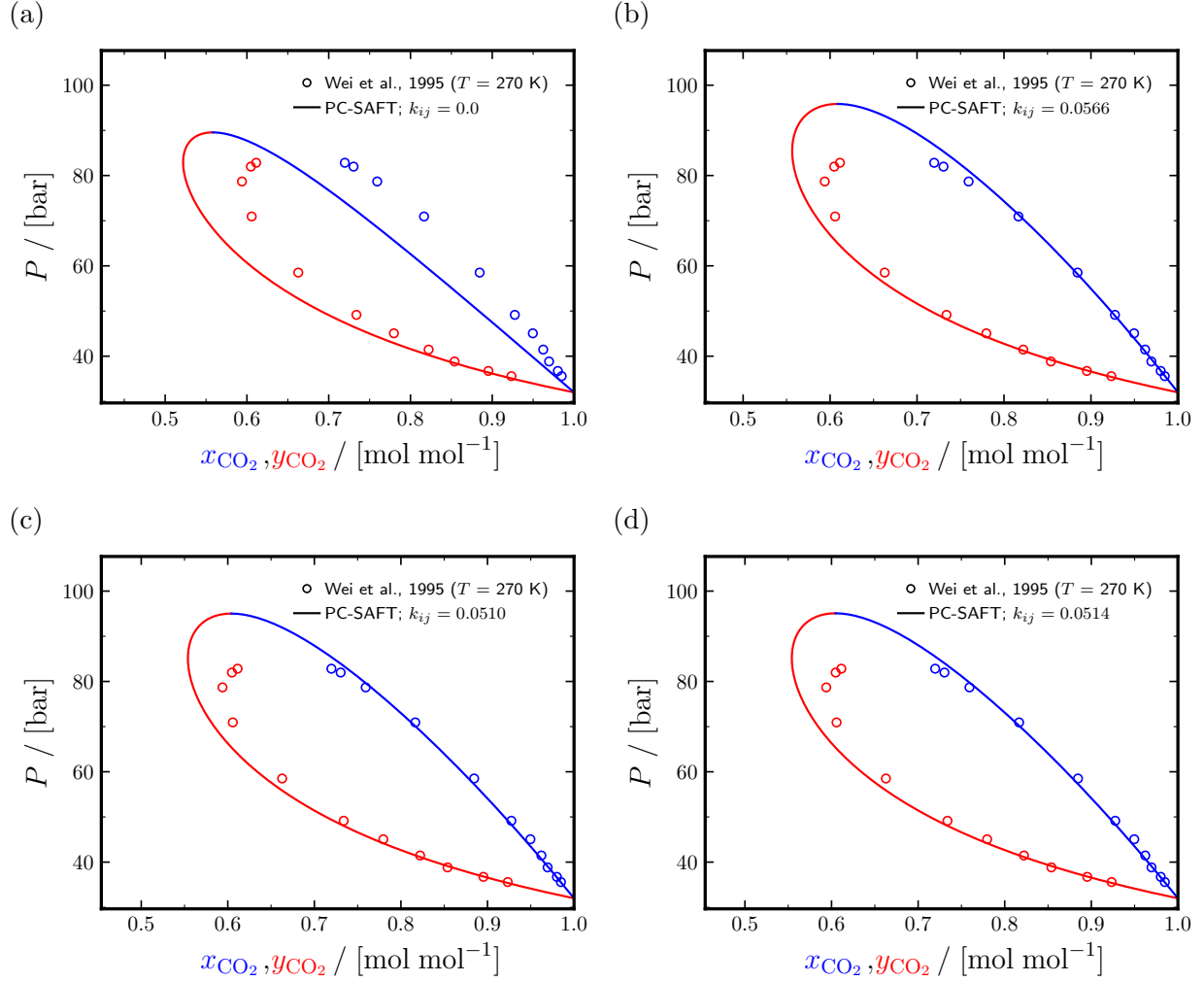


Fig. S5: Pxy diagrams of $\text{CO}_2\text{-CH}_4$ mixtures computed using the PC-SAFT EoS^{20,21} at 270 K, in comparison with the experimental data of Wei et al. (1995).²⁹ The diagrams were generated using (a) $k_{ij} = 0$, (b) $k_{ij} = 0.0566$, (c) $k_{ij} = 0.0510$, and (d) $k_{ij} = 0.0514$, where the values in (b)–(d) were obtained from cost-function optimizations performed using the experimental measurements²⁹ with distance, chemical potential, and saturation pressure criteria, respectively.

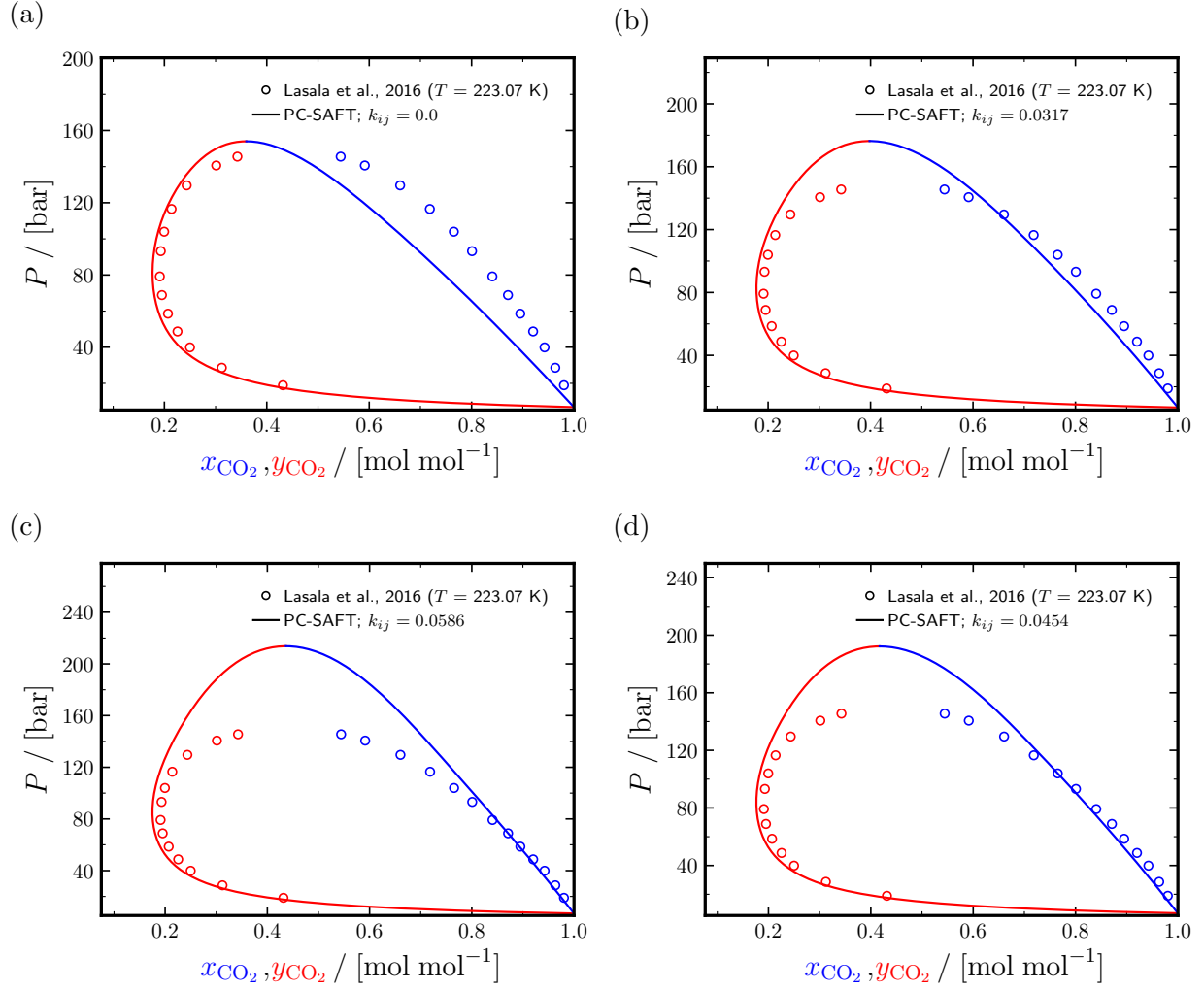


Fig. S6: Pxy diagrams of CO_2 -Ar mixtures computed using the PC-SAFT EoS^{20,21} at 223.07 K, in comparison with the experimental data of Lasala et al. (2016).³¹ The diagrams were generated using (a) $k_{ij} = 0$, (b) $k_{ij} = 0.0317$, (c) $k_{ij} = 0.0586$, and (d) $k_{ij} = 0.0454$, where the values in (b)–(d) were obtained from cost-function optimizations performed using the experimental measurements³¹ with distance, chemical potential, and saturation pressure criteria, respectively.

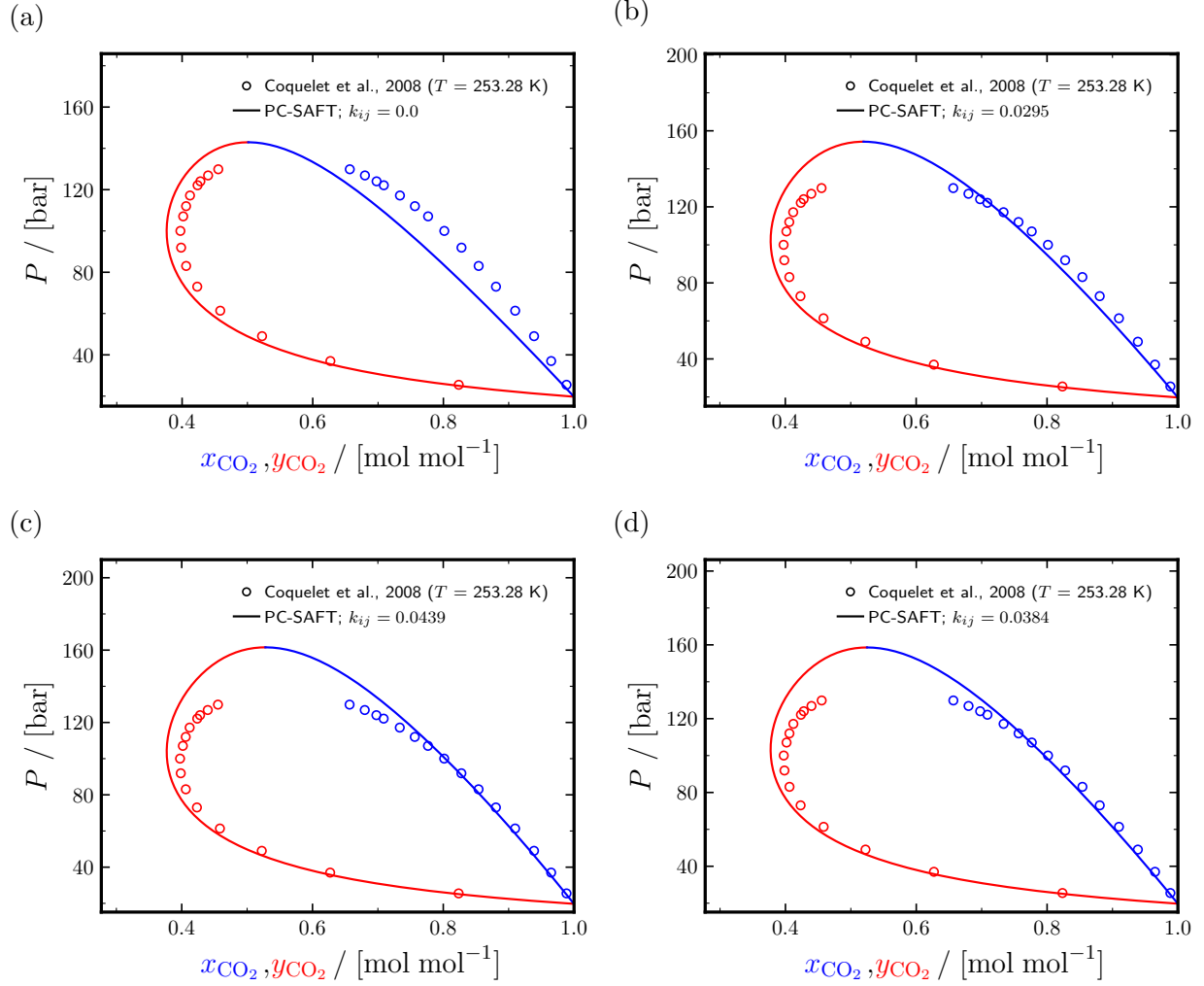


Fig. S7: Pxy diagrams of CO_2 -Ar mixtures computed using the PC-SAFT EoS^{20,21} at 253.28 K, in comparison with the experimental data of Coquelet et al. (2008).³⁰ The diagrams were generated using (a) $k_{ij} = 0$, (b) $k_{ij} = 0.0295$, (c) $k_{ij} = 0.0439$, and (d) $k_{ij} = 0.0384$, where the values in (b)–(d) were obtained from cost-function optimizations performed using the experimental measurements³⁰ with distance, chemical potential, and saturation pressure criteria, respectively.

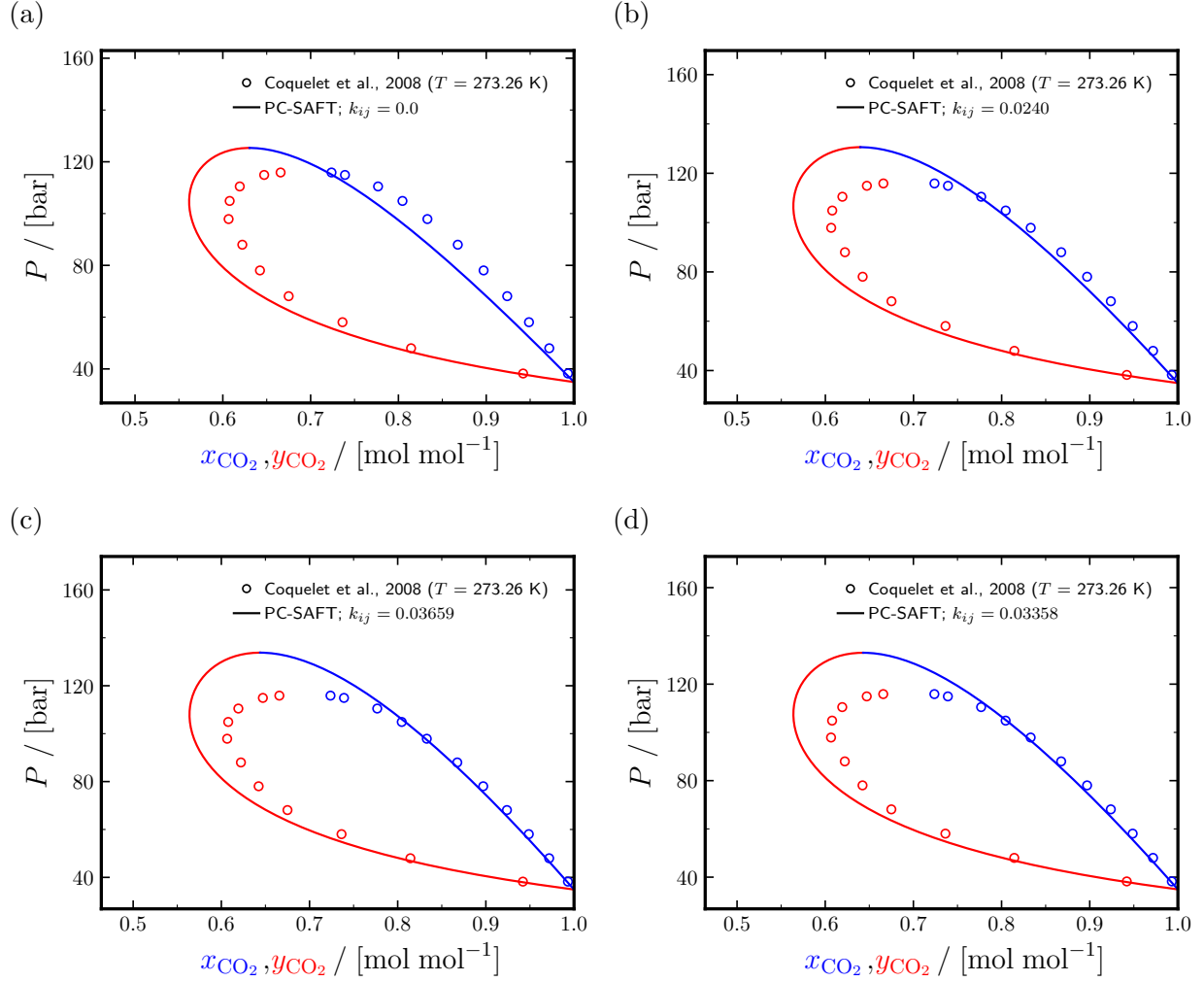


Fig. S8: Pxy diagrams of CO_2 -Ar mixtures computed using the PC-SAFT EoS^{20,21} at 273.26 K, in comparison with the experimental data of Coquelet et al. (2008).³⁰ The diagrams were generated using (a) $k_{ij} = 0$, (b) $k_{ij} = 0.0240$, (c) $k_{ij} = 0.0360$, and (d) $k_{ij} = 0.0330$, where the values in (b)–(d) were obtained from cost-function optimizations performed using the experimental measurements³⁰ with distance, chemical potential, and saturation pressure criteria, respectively.

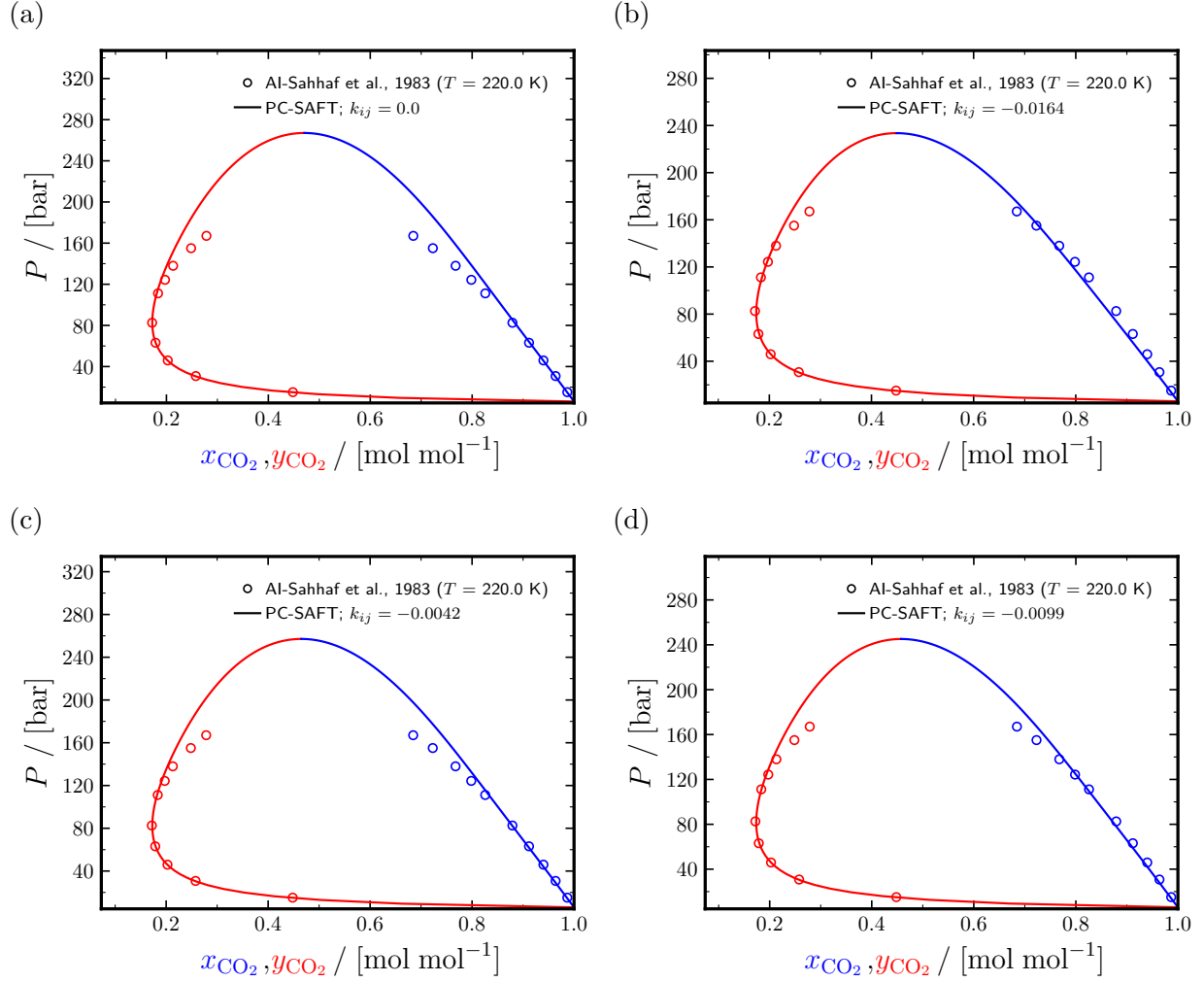


Fig. S9: Pxy diagrams of CO_2 - N_2 mixtures computed using the PC-SAFT EoS^{20,21} at 220 K, in comparison with the experimental data of Al-Sahhaf et al. (1983).³² The diagrams were generated using (a) $k_{ij} = 0$, (b) $k_{ij} = -0.0164$, (c) $k_{ij} = -0.0042$, and (d) $k_{ij} = -0.0099$, where the values in (b)–(d) were obtained from cost-function optimizations performed using the experimental measurements³² with distance, chemical potential, and saturation pressure criteria, respectively.

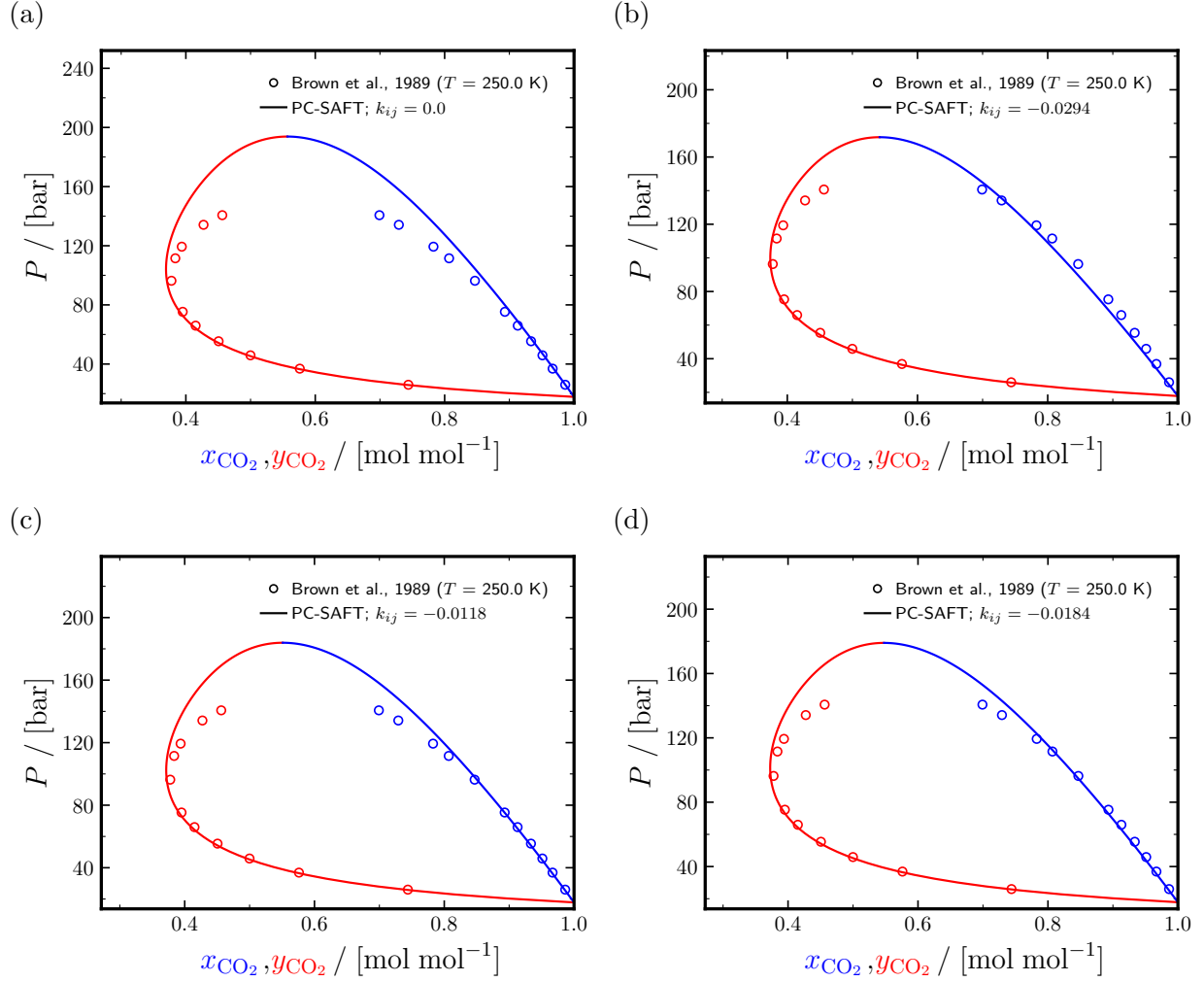


Fig. S10: Pxy diagrams of $\text{CO}_2\text{-N}_2$ mixtures computed using the PC-SAFT EoS^{20,21} at 250 K, in comparison with the experimental data of Brown et al. (1989).³³ The diagrams were generated using (a) $k_{ij} = 0$, (b) $k_{ij} = -0.0294$, (c) $k_{ij} = -0.00118$, and (d) $k_{ij} = -0.0184$, where the values in (b)–(d) were obtained from cost-function optimizations performed using the experimental measurements³³ with distance, chemical potential, and saturation pressure criteria, respectively.

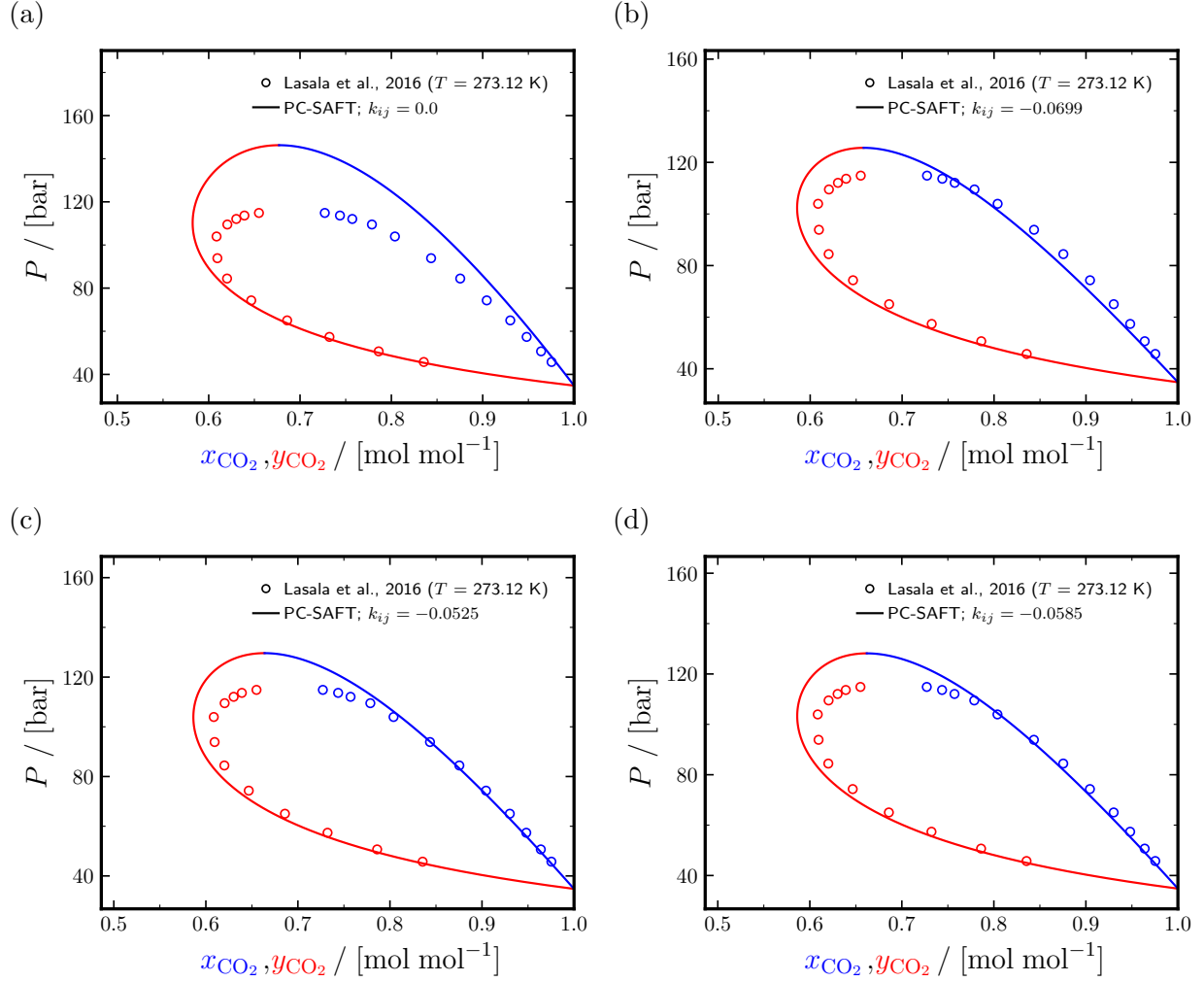


Fig. S11: Pxy diagrams of $\text{CO}_2\text{-N}_2$ mixtures computed using the PC-SAFT EoS^{20,21} at 273.12 K, in comparison with the experimental data of Lasala et al. (2016).³¹ The diagrams were generated using (a) $k_{ij} = 0$, (b) $k_{ij} = -0.0699$, (c) $k_{ij} = -0.0525$, and (d) $k_{ij} = -0.0585$, where the values in (b)–(d) were obtained from cost-function optimizations performed using the experimental measurements³¹ with distance, chemical potential, and saturation pressure criteria, respectively.

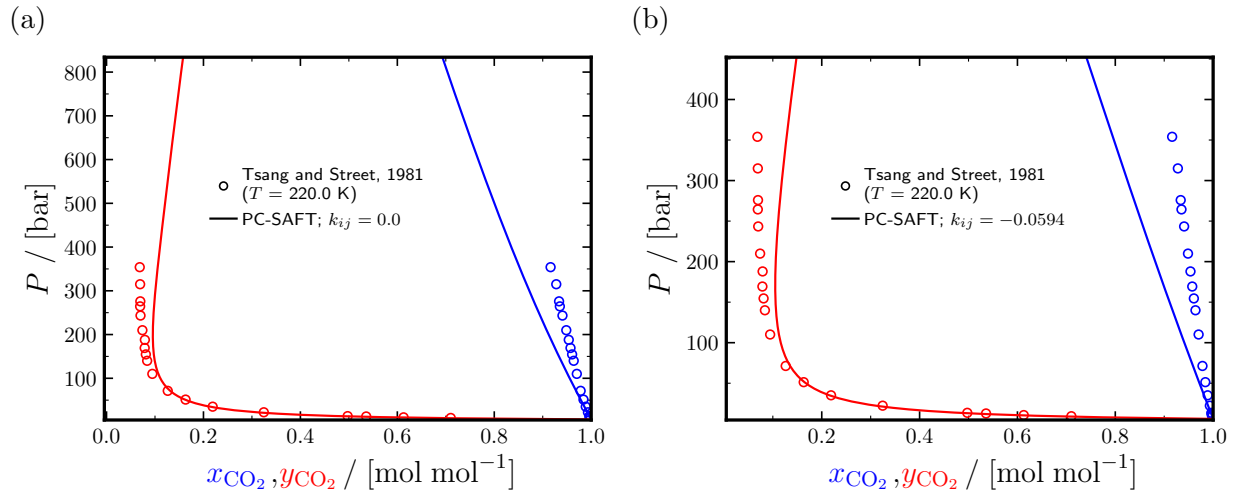


Fig. S12: Pxy diagrams of $\text{CO}_2\text{-H}_2$ mixtures computed using the PC-SAFT EoS^{20,21} at 220 K, in comparison with the experimental data of Tsang and Street (1981).³⁴ The diagrams were generated using (a) $k_{ij} = 0$ and (b) $k_{ij} = -0.0594$, where the value in (b) was obtained from a cost-function optimization performed using the experimental measurements³⁴ based on distance. The phase equilibria computations using the k_{ij} values optimized with the chemical potential and saturation pressure cost functions did not converge to the critical point.

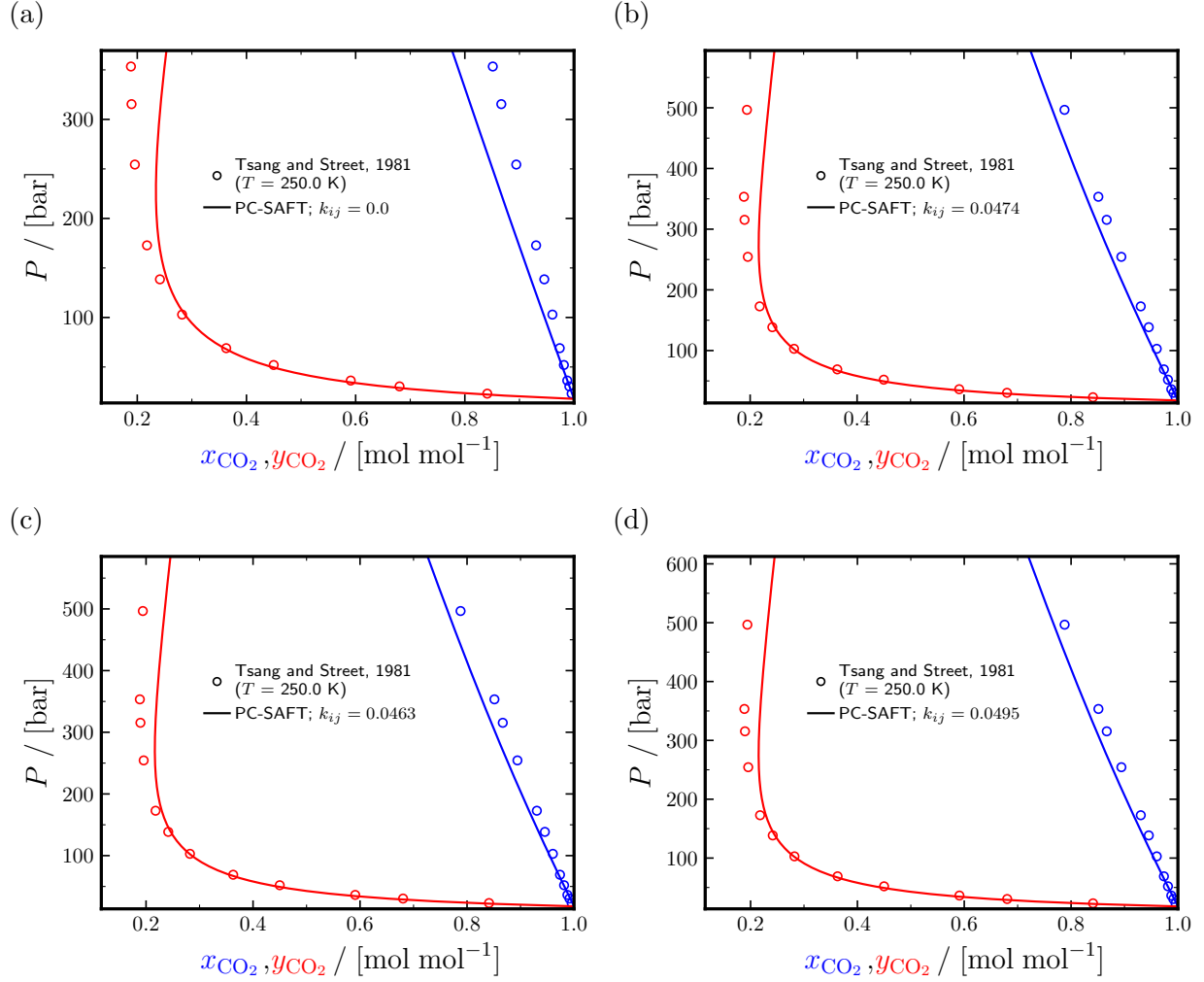


Fig. S13: Pxy diagrams of $\text{CO}_2\text{-H}_2$ mixtures computed using the PC-SAFT EoS^{20,21} at 250 K, in comparison with the experimental data of Tsang and Street (1981).³⁴ The diagrams were generated using (a) $k_{ij} = 0$, (b) $k_{ij} = 0.0474$, (c) $k_{ij} = 0.0463$, and (d) $k_{ij} = 0.0495$, where the values in (b)–(d) were obtained from cost-function optimizations performed using the experimental measurements³⁴ with distance, chemical potential, and saturation pressure criteria, respectively.

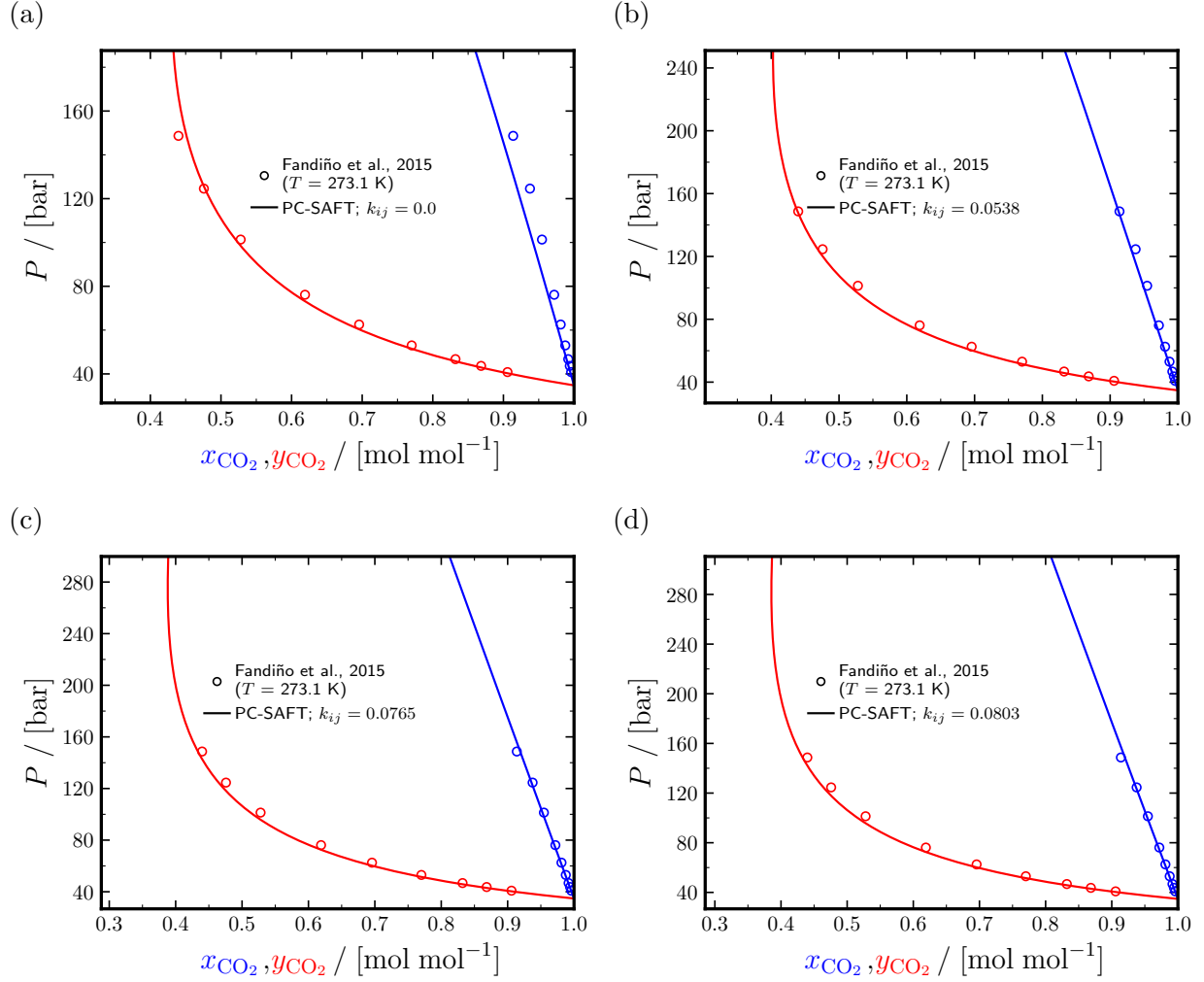


Fig. S14: Pxy diagrams of $\text{CO}_2\text{-H}_2$ mixtures computed using the PC-SAFT EoS^{20,21} at 273.1 K, in comparison with the experimental data of Fandiño et al. (2015).³⁵ The diagrams were generated using (a) $k_{ij} = 0$, (b) $k_{ij} = 0.0538$, (c) $k_{ij} = 0.0765$, and (d) $k_{ij} = 0.0803$, where the values in (b)–(d) were obtained from cost-function optimizations performed using the experimental measurements³⁵ with distance, chemical potential, and saturation pressure criteria, respectively.

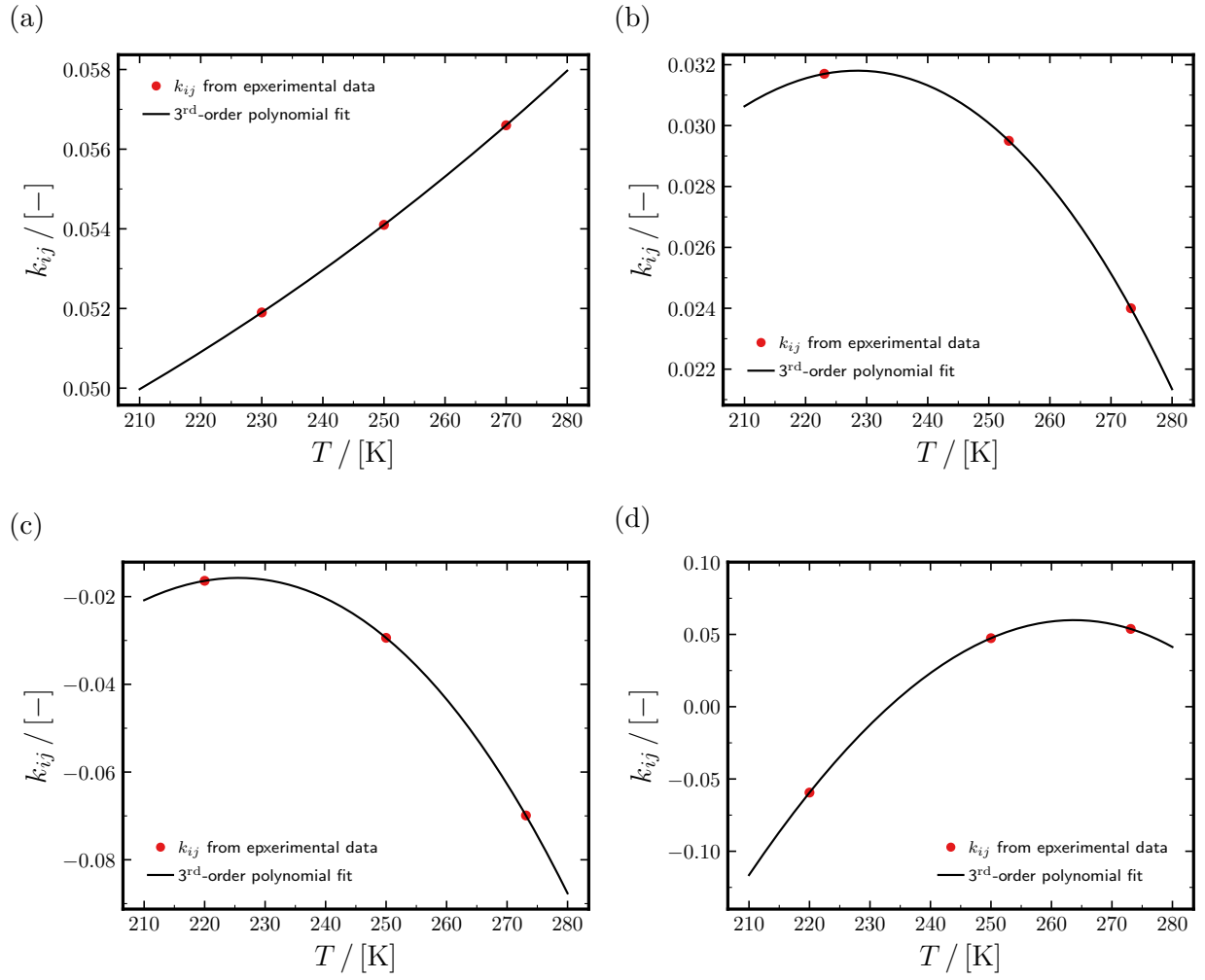


Fig. S15: Binary interaction parameters, k_{ij} s, fitted to a third-order polynomial using the k_{ij} values obtained by optimizing the distance-based cost function using experimental data^{29–35} for (a) $\text{CO}_2\text{-CH}_4$, (b) $\text{CO}_2\text{-Ar}$, (c) $\text{CO}_2\text{-N}_2$, and (d) $\text{CO}_2\text{-H}_2$ binary mixtures.

Table S24: Binary interaction parameters (k_{ij}) for CO₂-CH₄ at 220 K, CO₂-Ar at 220, 250, and 270 K, CO₂-N₂ at 270 K, and CO₂-H₂ at 270 K, obtained from the third-order polynomial fit.

Binary mixture	T / [K]	k_{ij} / [-]
CO ₂ -CH ₄	220.0	0.0509
CO ₂ -Ar	220.0	0.0316
	250.0	0.0301
	270.0	0.0251
CO ₂ -N ₂	270.0	−0.0627
CO ₂ -H ₂	270.0	0.0572

S9 Interfacial properties of binary CO₂ mixtures

In this section, IFTs, interfacial thicknesses (L_{10}^{90}), and enrichment factors (E_2) of the Light Boiling Component (LBC) for the CO₂-CH₄, CO₂-Ar, CO₂-N₂, and CO₂-H₂ binary mixtures computed from MD simulations, the PC-SAFT EoS^{20,21} + cDFT with $k_{ij} = 0$ and $k_{ij} \neq 0$ are presented. The uncertainties of IFTs, interfacial thicknesses (L_{10}^{90}), and enrichment factors (E_2) computed from MD simulations are also presented. At each temperature and pressure, two independent simulations were performed, each starting from a different initial configuration. For each property, including the equilibrium mole fractions of CO₂ in the liquid and vapor phases (x and y), IFTs, L_{10}^{90} , and E_2 , the mean values were computed as the average of the two simulations. Uncertainties were calculated as the standard deviation of the two simulations. IFTs of CO₂-CH₄ at 230, 250, and 270 K, for CO₂-Ar at 223.07, 253.28, and 273.26 K, for CO₂-N₂ at 220, 250, and 273.12 K, and for CO₂-H₂ at 220, 250, and 273.12 K are shown in Fig. S16. Similarly, interfacial thicknesses (L_{10}^{90}), and enrichment factors (E_2) are shown in Fig. S17 and Fig. S18, respectively. For the CO₂-CH₄ mixture, the experimental IFTs reported by Schenk et al.³⁷ at 233 K are compared with the IFTs computed at 230 K using MD simulations and the PC-SAFT EoS^{20,21} + cDFT with both $k_{ij} = 0$ and $k_{ij} \neq 0$.

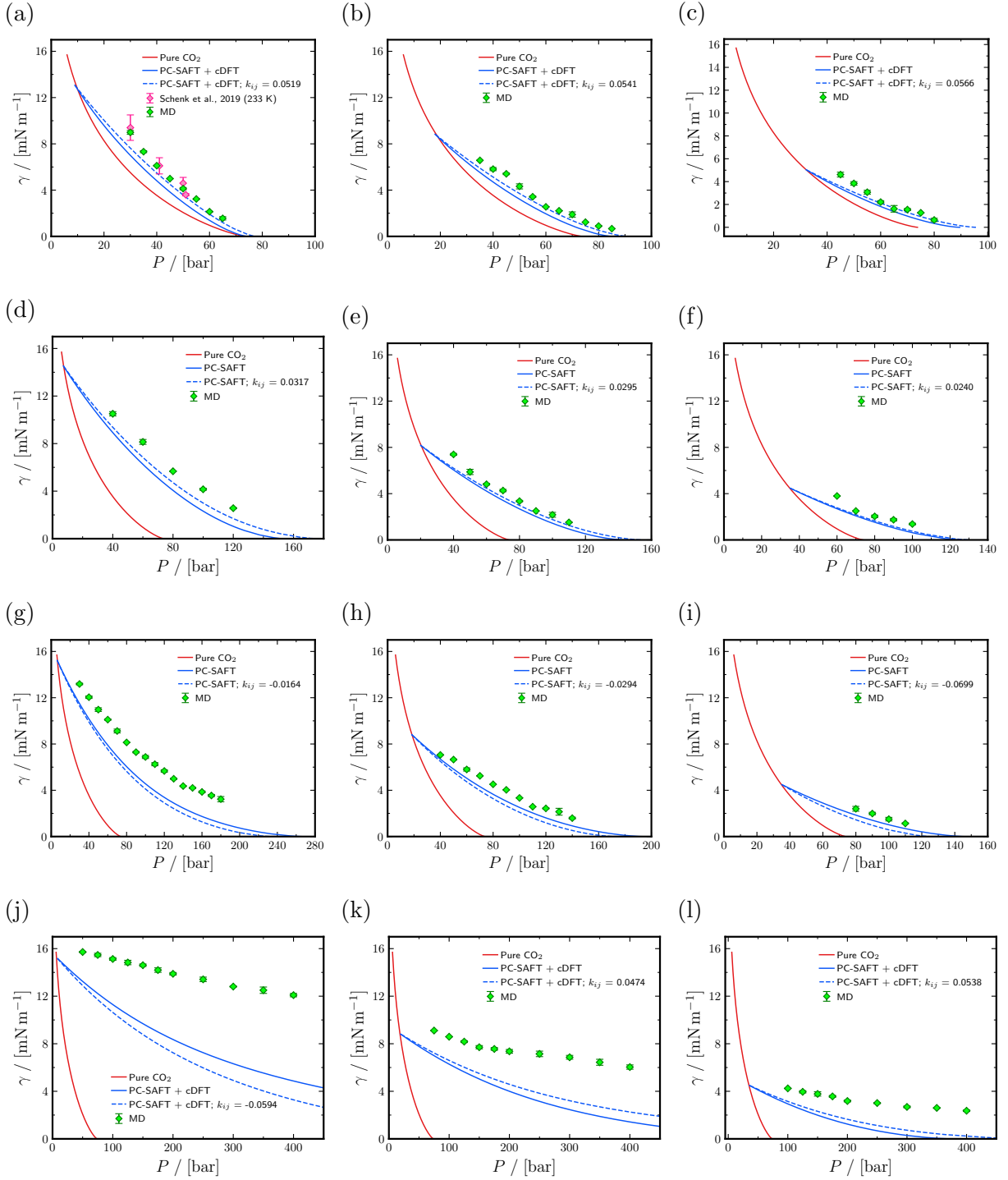


Fig. S16: Comparison of IFTs computed from MD simulations and from the PC-SAFT EoS^{20,21} + cDFT with $k_{ij} = 0$ and $k_{ij} \neq 0$ for CO₂-CH₄ mixtures at (a) 230 K, (b) 250 K, and (c) 270 K, CO₂-Ar mixtures at (d) 223.07 K, (e) 253.28 K, and (f) 273.26 K, CO₂-N₂ at (g) 220 K, (h) 250 K, and (i) 273.12 K and, CO₂-H₂ mixtures at (j) 220 K, (k) 250 K, and (l) 273.1 K.

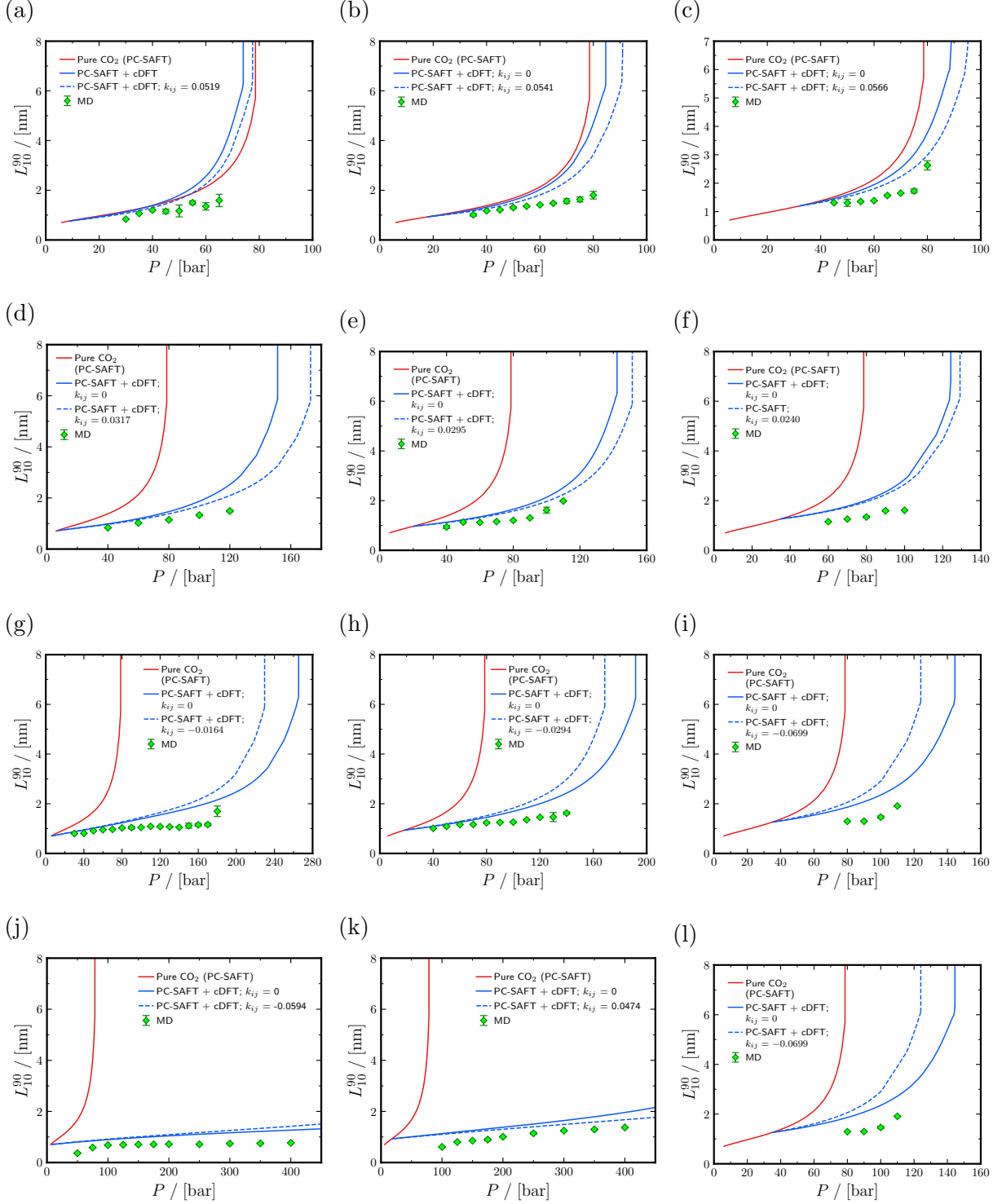


Fig. S17: Comparison of interfacial thicknesses (L_{10}^{90}) computed from MD simulations and from the PC-SAFT EoS^{20,21} + cDFT with $k_{ij} = 0$ and $k_{ij} \neq 0$ for $\text{CO}_2\text{-CH}_4$ mixtures at (a) 230 K, (b) 250 K, and (c) 270 K, $\text{CO}_2\text{-Ar}$ mixtures at (d) 223.07 K, (e) 253.28 K, and (f) 273.26 K, $\text{CO}_2\text{-N}_2$ at (g) 220 K, (h) 250 K, and (i) 273.12 K and, $\text{CO}_2\text{-H}_2$ mixtures at (j) 220 K, (k) 250 K, and (l) 273.1 K.

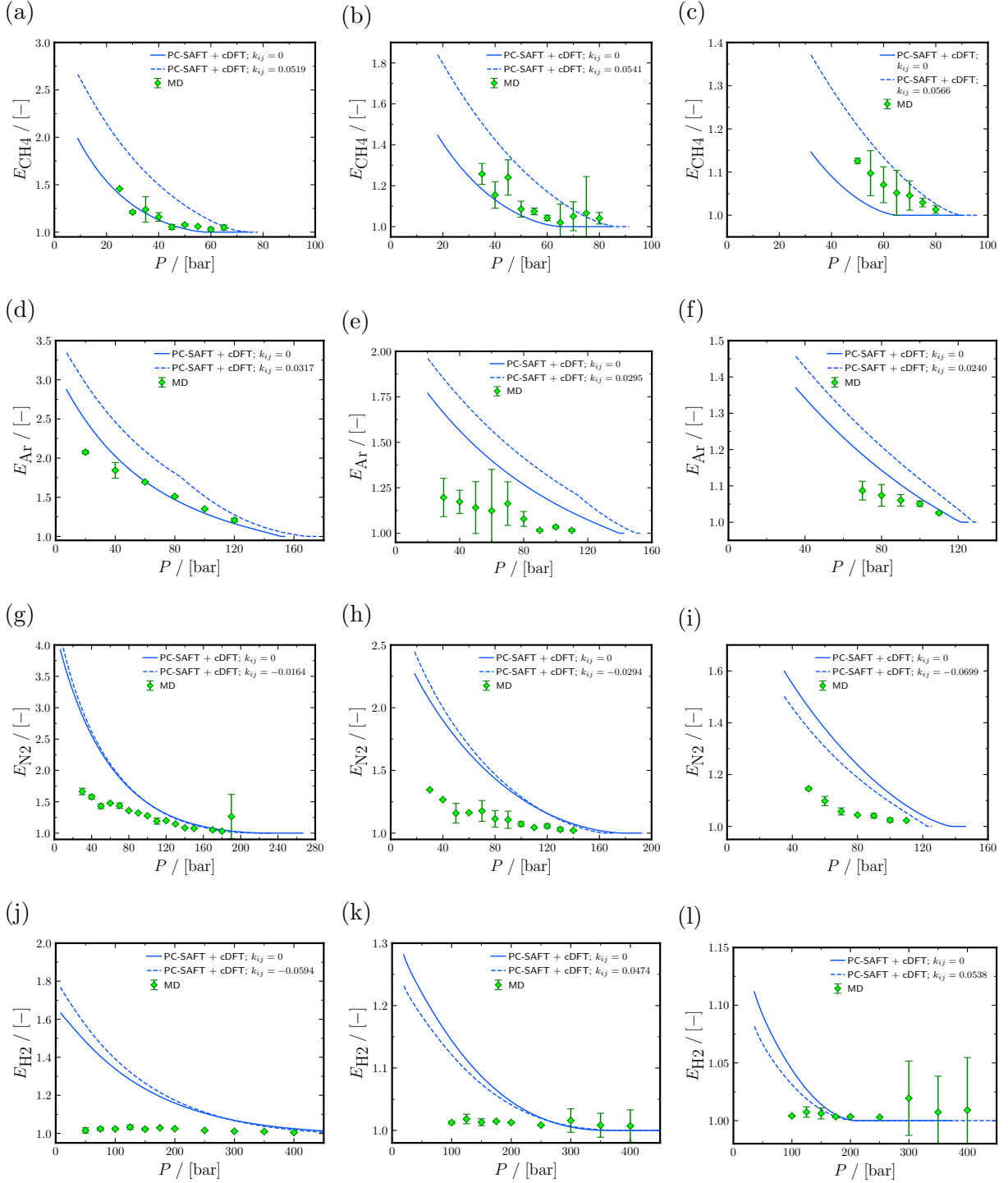


Fig. S18: Comparison of interfacial enrichment of the Light Boiling Component (LBC) (E_2) computed from MD simulations and from the PC-SAFT EoS^{20,21} + cDFT with $k_{ij} = 0$ and $k_{ij} \neq 0$ for CO₂-CH₄ mixtures at (a) 230 K, (b) 250 K, and (c) 270 K, CO₂-Ar mixtures at (d) 223.07 K, (e) 253.28 K, and (f) 273.26 K, CO₂-N₂ at (g) 220 K, (h) 250 K, and (i) 273.12 K and, CO₂-H₂ mixtures at (j) 220 K, (k) 250 K, and (l) 273.1 K.

S10 Phase equilibria and interfacial properties data of binary CO₂ mixtures

The raw Pxy data, IFTs, interfacial thicknesses (L_{10}^{90}), and enrichment factors (E_2) at the LBC for CO₂-CH₄ at 230, 250, and 270 K, for CO₂-Ar at 223.07, 253.28, and 273.26 K, for CO₂-N₂ at 220, 250, and 273.12 K, and for CO₂-H₂ at 220, 250, and 273.12 K, computed using the PC-SAFT EoS^{20,21} combined with cDFT with both $k_{ij} = 0$ and $k_{ij} \neq 0$, are provided in the accompanying .csv files listed in Table S25.

The raw Pxy data, IFTs, interfacial thicknesses (L_{10}^{90}), and enrichment factors (E_2) of the LBC are presented for the CO₂-CH₄, CO₂-Ar, CO₂-N₂, and CO₂-H₂ binary mixtures, together with their associated uncertainties. Results are shown for CO₂-CH₄ at 230, 250, and 270 K, for CO₂-Ar at 223.07, 253.28, and 273.26 K, for CO₂-N₂ at 220, 250, and 273.12 K, and for CO₂-H₂ at 220, 250, and 273.12 K. At each temperature and pressure, two independent simulations were performed, each starting from a different initial configuration. For each property, including the equilibrium mole fractions of CO₂ in the liquid and vapor phases (x and y), the IFTs, L_{10}^{90} , and E_2 , the mean values were computed as the average of the two simulations. For the liquid and vapor mole fractions of CO₂ (x and y), we report uncertainties that contain at least four significant digits. For IFTs, L_{10}^{90} , and E_2 , we report uncertainties with at least two significant digits.

Table S25: List of .csv files containing raw Pxy , IFTs, L_{10}^{90} , and E_2 data computed using PC-SAFT EoS^{20,21} + cDFT with $k_{ij} = 0$ and $k_{ij} \neq 0$.

Binary mixture	T / [K]	File name ($k_{ij} = 0$)	File name ($k_{ij} \neq 0$)
CO ₂ -CH ₄	230.0	IP_C02_CH4_kij=0_230K.csv	IP_C02_CH4_kij=0.0519_230K.csv
	250.0	IP_C02_CH4_kij=0_250K.csv	IP_C02_CH4_kij=0.0541_250K.csv
	270.0	IP_C02_CH4_kij=0_270K.csv	IP_C02_CH4_kij=0.0566_270K.csv
CO ₂ -Ar	223.0	IP_C02_Ar_kij=0_223K.csv	IP_C02_Ar_kij=0.0317_223K.csv
	253.3	IP_C02_Ar_kij=0_253K.csv	IP_C02_Ar_kij=0.0295_253K.csv
	273.3	IP_C02_Ar_kij=0_273K.csv	IP_C02_Ar_kij=0.0240_273K.csv
CO ₂ -N ₂	220.0	IP_C02_N2_kij=0_220K.csv	IP_C02_N2_kij=-0.0164_220K.csv
	250.0	IP_C02_N2_kij=0_250K.csv	IP_C02_N2_kij=-0.0294_250K.csv
	273.12	IP_C02_N2_kij=0_273K.csv	IP_C02_N2_kij=-0.0699_273K.csv
CO ₂ -H ₂	220.0	IP_C02_H2_kij=0_220K.csv	IP_C02_H2_kij=-0.0594_220K.csv
	250.0	IP_C02_H2_kij=0_250K.csv	IP_C02_H2_kij=0.0474_250K.csv
	273.12	IP_C02_H2_kij=0_273K.csv	IP_C02_H2_kij=0.0538_273K.csv

Table S26: The liquid and vapor mole fractions (x and y) of CO₂, interfacial tension (IFTs), interfacial thicknesses (L_{10}^{90}), and enrichment factors (E_{CH_4}) for the CO₂-CH₄ mixtures computed using MD simulations at 230 K.

P / [bar]	x_{CO_2} / [-]	y_{CO_2} / [-]	γ / [mN m ⁻¹]	L_{10}^{90} / [nm]	E_{CH_4} / [-]
30.0	0.8975 ± 0.0035	0.3990 ± 0.0035	8.99 ± 0.17	0.840	1.21 ± 0.02
35.0	0.859	0.340	7.32 ± 0.09	1.070	1.24 ± 0.13
40.0	0.8262 ± 0.0002	0.3036 ± 0.0015	6.11 ± 0.04	1.20 ± 0.01	1.16 ± 0.04
45.0	0.7859 ± 0.0064	0.2804 ± 0.0025	4.99 ± 0.01	1.15 ± 0.09	1.05 ± 0.03
50.0	0.7382 ± 0.0008	0.2562 ± 0.0070	4.13 ± 0.10	1.17 ± 0.24	1.08 ± 0.01
55.0	0.7018 ± 0.0117	0.2589 ± 0.0041	3.23 ± 0.05	1.50 ± 0.08	1.060
60.0	0.6364 ± 0.0015	0.2442 ± 0.0022	2.14 ± 0.03	1.35 ± 0.15	1.03 ± 0.02
65.0	0.6058 ± 0.0042	0.2870 ± 0.0471	1.56 ± 0.14	1.59 ± 0.25	1.05 ± 0.03

Table S27: The liquid and vapor mole fractions (x and y) of CO₂, interfacial tension (IFTs), interfacial thicknesses (L_{10}^{90}), and enrichment factors (E_{CH_4}) for the CO₂-CH₄ mixtures computed using MD simulations at 250 K.

P / [bar]	x_{CO_2} / [-]	y_{CO_2} / [-]	γ / [mN m ⁻¹]	L_{10}^{90} / [nm]	E_{CH_4} / [-]
35.0	0.912	0.571	6.570	1.01 ± 0.05	1.26 ± 0.05
40.0	0.8992 ± 0.0016	0.5451 ± 0.0053	5.82 ± 0.14	1.170	1.15 ± 0.06
45.0	0.8778 ± 0.0036	0.5120 ± 0.0031	5.41 ± 0.07	1.21 ± 0.01	1.24 ± 0.09
50.0	0.8455 ± 0.0045	0.4844 ± 0.0158	4.32 ± 0.22	1.30 ± 0.04	1.09 ± 0.04
55.0	0.8162 ± 0.0051	0.4423 ± 0.0083	3.43 ± 0.06	1.36 ± 0.02	1.07 ± 0.02
60.0	0.7797 ± 0.0024	0.4268 ± 0.0164	2.550	1.41 ± 0.02	1.04 ± 0.01
65.0	0.738	0.394	2.210	1.48 ± 0.03	1.02 ± 0.09
70.0	0.6901 ± 0.0037	0.4022 ± 0.0028	1.89 ± 0.23	1.56 ± 0.10	1.05 ± 0.07
75.0	0.631	0.386	1.230	1.63 ± 0.10	1.07 ± 0.18
80.0	0.5984 ± 0.0249	0.4006 ± 0.0039	0.89 ± 0.03	1.80 ± 0.15	1.04 ± 0.03

Table S28: The liquid and vapor mole fractions (x and y) of CO₂, interfacial tension (IFTs), interfacial thicknesses (L_{10}^{90}), and enrichment factors (E_{CH_4}) for the CO₂-CH₄ mixtures computed using MD simulations at 270 K.

P / [bar]	x_{CO_2} / [-]	y_{CO_2} / [-]	γ / [mN m ⁻¹]	L_{10}^{90} / [nm]	E_{CH_4} / [-]
50.0	0.915	0.721	3.84 ± 0.15	1.31 ± 0.13	1.13 ± 0.01
55.0	0.9070 ± 0.0018	0.7006 ± 0.0037	3.07 ± 0.17	1.35 ± 0.01	1.10 ± 0.05
60.0	0.8779 ± 0.0044	0.6612 ± 0.0211	2.21 ± 0.12	1.380	1.07 ± 0.04
65.0	0.8597 ± 0.0008	0.6560 ± 0.0096	1.62 ± 0.29	1.57 ± 0.03	1.05 ± 0.05
70.0	0.8191 ± 0.0061	0.6056 ± 0.0076	1.55 ± 0.09	1.65 ± 0.01	1.05 ± 0.03
75.0	0.7966 ± 0.0071	0.6308 ± 0.0221	1.27 ± 0.01	1.72 ± 0.07	1.03 ± 0.01
80.0	0.7488 ± 0.0049	0.6454 ± 0.0145	0.64 ± 0.15	2.63 ± 0.16	1.01 ± 0.01

Table S29: The liquid and vapor mole fractions (x and y) of CO₂, interfacial tension (IFTs), interfacial thicknesses (L_{10}^{90}), and enrichment factors (E_{Ar}) for the CO₂-Ar mixtures computed using MD simulations at 223.07 K.

P / [bar]	x_{CO_2} / [-]	y_{CO_2} / [-]	γ / [mN m ⁻¹]	L_{10}^{90} / [nm]	E_{Ar} / [-]
40.0	0.9383 ± 0.0038	0.2556 ± 0.0125	10.51 ± 0.18	0.84 ± 0.01	1.84 ± 0.10
60.0	0.8961 ± 0.0040	0.2035 ± 0.0023	8.14 ± 0.20	1.020	1.70 ± 0.01
80.0	0.853	0.193	5.67 ± 0.01	1.140	1.51 ± 0.01
100.0	0.796	0.179	4.16 ± 0.08	1.33 ± 0.03	1.350
120.0	0.762	0.194	2.57 ± 0.05	1.49 ± 0.03	1.21 ± 0.02

Table S30: The liquid and vapor mole fractions (x and y) of CO₂, interfacial tension (IFTs), interfacial thicknesses (L_{10}^{90}), and enrichment factors (E_{Ar}) for the CO₂-Ar mixtures computed using MD simulations at 253.28 K.

P / [bar]	x_{CO_2} / [-]	y_{CO_2} / [-]	γ / [mN m ⁻¹]	L_{10}^{90} / [nm]	E_{Ar} / [-]
50.0	0.9332 ± 0.0014	0.5281 ± 0.0055	5.88 ± 0.23	1.130	1.14 ± 0.14
60.0	0.8915 ± 0.0106	0.478	4.81 ± 0.07	1.12 ± 0.01	1.12 ± 0.23
70.0	0.8670 ± 0.0050	0.4384 ± 0.0074	4.27 ± 0.10	1.160	1.16 ± 0.12
80.0	0.8512 ± 0.0112	0.4071 ± 0.0072	3.35 ± 0.06	1.200	1.08 ± 0.04
90.0	0.8130 ± 0.0052	0.4012 ± 0.0011	2.51 ± 0.01	1.310	1.02 ± 0.01
100.0	0.7943 ± 0.0185	0.3916 ± 0.0021	2.17 ± 0.21	1.62 ± 0.12	1.03 ± 0.01
110.0	0.7645 ± 0.0062	0.3838 ± 0.0199	1.52 ± 0.03	1.98 ± 0.03	1.02 ± 0.01

Table S31: The liquid and vapor mole fractions (x and y) of CO₂, interfacial tension (IFTs), interfacial thicknesses (L_{10}^{90}), and enrichment factors (E_{Ar}) for the CO₂-Ar mixtures computed using MD simulations at 273.26 K.

P / [bar]	x_{CO_2} / [-]	y_{CO_2} / [-]	γ / [mN m ⁻¹]	L_{10}^{90} / [nm]	E_{Ar} / [-]
70.0	0.9179 ± 0.0045	0.6758 ± 0.0174	2.300	1.260	1.09 ± 0.03
80.0	0.8967 ± 0.0047	0.6344 ± 0.0233	1.84 ± 0.11	1.34 ± 0.03	1.07 ± 0.03
90.0	0.863	0.597	1.54 ± 0.11	1.59 ± 0.02	1.06 ± 0.02
100.0	0.8284 ± 0.0062	0.6135 ± 0.0008	1.17 ± 0.07	1.61 ± 0.01	1.05 ± 0.01

Table S32: The liquid and vapor mole fractions (x and y) of CO₂, interfacial tension (IFTs), interfacial thicknesses (L_{10}^{90}), and enrichment factors (E_{N_2}) for the CO₂-N₂ mixtures computed using MD simulations at 220 K.

P / [bar]	x_{CO_2} / [-]	y_{CO_2} / [-]	γ / [mN m ⁻¹]	L_{10}^{90} / [nm]	E_{N_2} / [-]
30.0	0.9816 ± 0.0004	0.2744 ± 0.0129	13.19 ± 0.07	0.800	1.66 ± 0.05
40.0	0.9761 ± 0.0006	0.2385 ± 0.0210	12.05 ± 0.11	0.810	1.58 ± 0.03
50.0	0.9666 ± 0.0015	0.2058 ± 0.0049	10.97 ± 0.16	0.910	1.43 ± 0.04
60.0	0.9608 ± 0.0001	0.1864 ± 0.0002	10.11 ± 0.05	0.950	1.48 ± 0.01
70.0	0.9523 ± 0.0010	0.1736 ± 0.0010	9.12 ± 0.17	0.970	1.44 ± 0.04
80.0	0.9466 ± 0.0009	0.1759 ± 0.0006	8.14 ± 0.01	1.03 ± 0.02	1.36 ± 0.01
90.0	0.9364 ± 0.0003	0.1738 ± 0.0013	7.29 ± 0.06	1.04 ± 0.05	1.320
100.0	0.9311 ± 0.0127	0.1681 ± 0.0007	6.88 ± 0.13	1.050	1.28 ± 0.01
110.0	0.9252 ± 0.0018	0.1682 ± 0.0023	6.25 ± 0.14	1.09 ± 0.01	1.19 ± 0.05
120.0	0.9177 ± 0.0067	0.1675 ± 0.0049	5.67 ± 0.10	1.08 ± 0.02	1.20 ± 0.01
130.0	0.9082 ± 0.0053	0.1761 ± 0.0088	4.99 ± 0.06	1.07 ± 0.02	1.15 ± 0.01
140.0	0.8955 ± 0.0034	0.1868 ± 0.0020	4.36 ± 0.04	1.05 ± 0.01	1.080
150.0	0.8945 ± 0.0023	0.1831 ± 0.0015	4.20 ± 0.06	1.12 ± 0.10	1.080
170.0	0.8845 ± 0.0037	0.1846 ± 0.0028	3.55 ± 0.09	1.16 ± 0.05	1.05 ± 0.02
180.0	0.8779 ± 0.0034	0.1918 ± 0.0068	3.23 ± 0.23	1.70 ± 0.21	1.030

Table S33: The liquid and vapor mole fractions (x and y) of CO₂, interfacial tension (IFTs), interfacial thicknesses (L_{10}^{90}), and enrichment factors (E_{N_2}) for the CO₂-N₂ mixtures computed using MD simulations at 250 K.

P / [bar]	x_{CO_2} / [-]	y_{CO_2} / [-]	γ / [mN m ⁻¹]	L_{10}^{90} / [nm]	E_{N_2} / [-]
40.0	0.9780 ± 0.0008	0.6123 ± 0.0064	7.06 ± 0.02	1.010	1.270
50.0	0.9669 ± 0.0003	0.5082 ± 0.0057	6.65 ± 0.04	1.09 ± 0.01	1.16 ± 0.08
60.0	0.9561 ± 0.0011	0.4594 ± 0.0177	5.80 ± 0.14	1.160	1.160
70.0	0.9440 ± 0.0007	0.4290 ± 0.0054	5.250	1.170	1.18 ± 0.08
80.0	0.933	0.396	4.520	1.24 ± 0.03	1.11 ± 0.07
90.0	0.920	0.381	4.04 ± 0.01	1.25 ± 0.02	1.11 ± 0.07
100.0	0.9070 ± 0.0037	0.3830 ± 0.0032	3.34 ± 0.01	1.26 ± 0.02	1.07 ± 0.02
110.0	0.8882 ± 0.0023	0.3820 ± 0.0112	2.58 ± 0.01	1.35 ± 0.02	1.050
120.0	0.8782 ± 0.0023	0.3740 ± 0.0069	2.43 ± 0.04	1.45 ± 0.01	1.06 ± 0.02
130.0	0.8633 ± 0.0045	0.3725 ± 0.0234	2.14 ± 0.30	1.46 ± 0.18	1.03 ± 0.02
140.0	0.8456 ± 0.0072	0.3822 ± 0.0210	1.59 ± 0.08	1.62 ± 0.05	1.02 ± 0.01

Table S34: The liquid and vapor mole fractions (x and y) of CO₂, interfacial tension (IFTs), interfacial thicknesses (L_{10}^{90}), and enrichment factors (E_{N_2}) for the CO₂-N₂ mixtures computed using MD simulations at 273.26 K.

P / [bar]	x_{CO_2} / [-]	y_{CO_2} / [-]	γ / [mN m ⁻¹]	L_{10}^{90} / [nm]	E_{N_2} / [-]
80.0	0.9392 ± 0.0011	0.6480 ± 0.0013	2.40 ± 0.21	1.290	1.040
90.0	0.9212 ± 0.0071	0.6150 ± 0.0040	1.99 ± 0.11	1.30 ± 0.02	1.04 ± 0.01
100.0	0.8946 ± 0.0031	0.6005 ± 0.0035	1.50 ± 0.16	1.46 ± 0.05	1.02 ± 0.01
110.0	0.8804 ± 0.0059	0.5878 ± 0.0008	1.13 ± 0.05	1.91 ± 0.01	1.020

Table S35: The liquid and vapor mole fractions (x and y) of CO₂, interfacial tension (IFTs), interfacial thicknesses (L_{10}^{90}), and enrichment factors (E_{H_2}) for the CO₂-H₂ mixtures computed using MD simulations at 220 K.

P / [bar]	x_{CO_2} / [-]	y_{CO_2} / [-]	γ / [mN m ⁻¹]	L_{10}^{90} / [nm]	E_{H_2} / [-]
50.0	0.9908 ± 0.0018	0.1647 ± 0.0040	15.72 ± 0.04	0.370	1.02 ± 0.01
75.0	0.990	0.1154 ± 0.0004	15.47 ± 0.14	0.590	1.02 ± 0.01
100.0	0.9880 ± 0.0003	0.0908 ± 0.0006	15.13 ± 0.11	0.680	1.020
125.0	0.9852 ± 0.0003	0.0788 ± 0.0009	14.83 ± 0.19	0.700	1.03 ± 0.01
150.0	0.9825 ± 0.0006	0.0709 ± 0.0009	14.61 ± 0.10	0.710	1.02 ± 0.01
175.0	0.9794 ± 0.0001	0.0649 ± 0.0007	14.20 ± 0.19	0.710	1.030
200.0	0.9777 ± 0.0007	0.0600 ± 0.0005	13.88 ± 0.11	0.72 ± 0.01	1.020
250.0	0.9732 ± 0.0011	0.0546 ± 0.0003	13.42 ± 0.19	0.710	1.020
300.0	0.9678 ± 0.0015	0.0520 ± 0.0005	12.81 ± 0.04	0.740	1.010
350.0	0.9632 ± 0.0001	0.0488 ± 0.0013	12.50 ± 0.27	0.75 ± 0.01	1.010
400.0	0.959	0.047	12.09 ± 0.14	0.770	1.000

Table S36: The liquid and vapor mole fractions (x and y) of CO₂, interfacial tension (IFTs), interfacial thicknesses (L_{10}^{90}), and enrichment factors (E_{H_2}) for the CO₂-H₂ mixtures computed using MD simulations at 250 K.

P / [bar]	x_{CO_2} / [-]	y_{CO_2} / [-]	γ / [mN m ⁻¹]	L_{10}^{90} / [nm]	E_{H_2} / [-]
100.0	0.9704 ± 0.0101	0.2727 ± 0.0059	8.59 ± 0.01	0.61 ± 0.01	1.010
125.0	0.973	0.237	8.18 ± 0.08	0.80 ± 0.01	1.02 ± 0.01
150.0	0.9648 ± 0.0022	0.2192 ± 0.0039	7.71 ± 0.14	0.850	1.01 ± 0.01
175.0	0.963	0.181	7.56 ± 0.12	0.90 ± 0.04	1.010
200.0	0.956	0.183	7.36 ± 0.15	1.01 ± 0.02	1.010
250.0	0.942	0.161	7.15 ± 0.23	1.14 ± 0.04	1.010
300.0	0.926	0.135	6.86 ± 0.16	1.25 ± 0.02	1.02 ± 0.02
350.0	0.930	0.136	6.44 ± 0.25	1.30 ± 0.01	1.01 ± 0.02
400.0	0.920	0.136	6.04 ± 0.18	1.37 ± 0.01	1.01 ± 0.03

Table S37: The liquid and vapor mole fractions (x and y) of CO₂, interfacial tension (IFTs), interfacial thicknesses (L_{10}^{90}), and enrichment factors (E_{H_2}) for the CO₂-H₂ mixtures computed using MD simulations at 273.1 K.

P / [bar]	x_{CO_2} / [-]	y_{CO_2} / [-]	γ / [mN m ⁻¹]	L_{10}^{90} / [nm]	E_{H_2} / [-]
100.0	0.9685 ± 0.0029	0.5439 ± 0.0094	4.24 ± 0.01	0.56 ± 0.02	1.000
125.0	0.9635 ± 0.0006	0.4629 ± 0.0044	3.96 ± 0.10	0.88 ± 0.01	1.010
150.0	0.9547 ± 0.0004	0.4100 ± 0.0082	3.78 ± 0.18	0.97 ± 0.01	1.010
175.0	0.9422 ± 0.0016	0.3693 ± 0.0093	3.57 ± 0.07	1.04 ± 0.06	1.000
200.0	0.9339 ± 0.0002	0.3634 ± 0.0055	3.18 ± 0.07	1.23 ± 0.03	1.000
250.0	0.9158 ± 0.0007	0.3147 ± 0.0004	3.01 ± 0.05	1.47 ± 0.07	1.000
300.0	0.8853 ± 0.0001	0.2999 ± 0.0006	2.69 ± 0.07	1.63 ± 0.03	1.02 ± 0.03
350.0	0.8736 ± 0.0006	0.2755 ± 0.0046	2.60 ± 0.03	1.71 ± 0.01	1.01 ± 0.03
400.0	0.8621 ± 0.0010	0.2660 ± 0.0074	2.36 ± 0.06	1.84 ± 0.02	1.01 ± 0.05

S11 Superheat limit (SHL) and Supercool limit (SCL) data of pure CO₂ and binary CO₂ mixtures

The raw Superheat limit (SHL) and Supercool limit (SCL) of pure CO₂ computed using the Classical Nucleation Theory (CNT)³⁸ are shown in this section. The SHL and SCL of pure CO₂ were computed using IFTs obtained from the PC-SAFT EoS^{20,21} combined with cDFT, Guggenheim–Katayama equation,¹³ and MD simulations. In addition, the SHL and SCL of pure CO₂ reported by Aursand et al.³⁹ and Hammer et al.,⁴⁰ respectively, were obtained using the WebPlotDigitalizer software⁴¹ are also presented in this section.

The raw SHL, SCL and saturation lines (dew and bubble lines) of CO₂-CH₄, CO₂-Ar, CO₂-N₂, and CO₂-H₂ binary mixtures containing 95 mol% CO₂ and 5 mol% impurities (CH₄, Ar, N₂, and H₂) are provided in the accompanying .csv files listed in Table S44. For CO₂-CH₄, CO₂-Ar, and CO₂-N₂ mixtures, the SHL and SCL were computed using IFTs values obtained from the Parachor model.⁴² For CO₂-H₂ mixtures, the SHL and SCL were computed using the IFTs values obtained using the Winterfeld Scriven Davis (WSD) model.⁴³

Table S38: The Superheat limit (SHL) of pure CO₂, extracted from the plot presented by Aursand et al..³⁹

T / [K]	SHL / [bar]
269.83	0.96
271.25	4.30
272.54	8.12
273.97	11.47
275.40	14.81
276.97	18.63
278.11	21.02
279.68	24.37
280.97	28.19

Table S38: The Superheat limit (SHL) of pure CO₂, extracted from the plot presented by Aursand et al..³⁹

T / [K]	SHL / [bar]
282.25	30.58
283.83	34.64
285.04	37.03
286.47	40.14
287.75	42.76
289.18	45.87
290.61	48.50
292.04	51.37
293.47	54.23
294.75	57.10
296.11	59.49
297.54	62.35
298.89	64.51
300.25	66.42
301.75	69.04
303.10	71.43

Table S39: The Supercool limit (SCL) of pure CO₂, extracted from the plot presented by Hammer et al..⁴⁰

T / [K]	SCL / [bar]
233.24	1.48
235.55	1.59
237.76	1.69
239.39	1.76

Table S39: The Supercool limit (SCL) of pure CO₂, extracted from the plot presented by Hammer et al..⁴⁰

T / [K]	SCL / [bar]
241.20	1.86
242.86	1.93
244.58	2.01
246.11	2.10
247.10	2.15
248.39	2.22
249.68	2.30
250.97	2.37
252.44	2.45
254.10	2.54
255.58	2.63
257.24	2.74
258.83	2.84
259.94	2.91
261.05	2.98
262.27	3.05
263.81	3.16
265.10	3.25
266.58	3.36
267.99	3.46
269.28	3.56
270.69	3.67
272.35	3.80
273.95	3.92

Table S39: The Supercool limit (SCL) of pure CO₂, extracted from the plot presented by Hammer et al..⁴⁰

T / [K]	SCL / [bar]
275.42	4.06
276.59	4.15
277.94	4.27
279.14	4.39
280.46	4.51
281.63	4.62
282.98	4.76
284.15	4.88
285.56	5.02
287.34	5.20
289.43	5.43
290.97	5.59
292.32	5.76
293.61	5.90
295.15	6.08
296.38	6.24
297.73	6.41
298.77	6.56
299.88	6.71
301.17	6.87
302.27	7.03
303.56	7.22
304.95	7.41
305.87	7.57

Table S39: The Supercool limit (SCL) of pure CO₂, extracted from the plot presented by Hammer et al..⁴⁰

T / [K]	SCL / [bar]
306.76	7.71

Table S40: The Superheat limit (SHL) of pure CO₂ computed from IFTs estimated using the PC-SAFT EoS^{20,21} combined with cDFT.³⁸ The liquid spinodal limit of pure CO₂, computed using the PC-SAFT EoS.^{20,21} Values of the SHL and spinodal less than zero for the corresponding temperatures are indicated as ‘-’.

T / [K]	SHL / [bar]	Liquid spinodal / [bar]
270.0	1.63	-
271.0	4.14	-
272.0	6.61	-
272.0	6.61	-
273.0	9.06	-
274.0	11.48	-
275.0	13.87	-
276.0	16.23	-
277.0	18.57	-
278.0	20.88	-
279.0	23.16	-
280.0	25.42	-
281.0	27.65	-
282.0	29.86	-
283.0	32.04	-
284.0	34.20	1.93
285.0	36.34	6.08
286.0	38.45	10.16

Table S40: The Superheat limit (SHL) of pure CO₂ computed from IFTs estimated using the PC-SAFT EoS^{20,21} combined with cDFT.³⁸ The liquid spinodal limit of pure CO₂, computed using the PC-SAFT EoS.^{20,21} Values of the SHL and spinodal less than zero for the corresponding temperatures are indicated as ‘-’.

T / [K]	SHL / [bar]	Liquid spinodal / [bar]
287.0	40.54	14.17
288.0	42.60	18.11
289.0	44.65	21.98
290.0	46.67	25.78
291.0	48.68	29.50
292.0	50.66	33.15
293.0	52.63	36.73
294.0	54.57	40.23
295.0	56.50	43.65
296.0	58.41	47.00
297.0	60.30	50.27
298.0	62.18	53.46
299.0	64.04	56.57
300.0	65.89	59.60
301.0	67.73	62.54
302.0	69.55	65.39
303.0	71.36	68.16
304.0	73.17	70.82
305.0	74.96	73.39
306.0	76.75	75.85
307.0	78.54	78.18
309.0	82.11	82.38

Table S41: The Supercool limit (SCL) of pure CO₂ computed from IFTs estimated using the PC-SAFT EoS^{20,21} combined with cDFT.³⁸ The vapor spinodal limit of pure CO₂, computed using the PC-SAFT EoS.^{20,21}

T / [K]	SCL / [bar]	Liquid spinodal / [bar]
220.0	8.92	23.91
221.0	9.20	24.24
222.0	9.48	24.57
223.0	9.78	24.91
224.0	10.08	25.25
225.0	10.39	25.60
226.0	10.70	25.95
227.0	11.02	26.31
228.0	11.35	26.67
229.0	11.69	27.03
230.0	12.04	27.40
231.0	12.39	27.77
232.0	12.75	28.14
233.0	13.12	28.52
234.0	13.50	28.91
235.0	13.89	29.30
236.0	14.28	29.69
237.0	14.69	30.09
238.0	15.10	30.49
239.0	15.52	30.90
240.0	15.95	31.31
241.0	16.40	31.73
242.0	16.85	32.15

Table S41: The Supercool limit (SCL) of pure CO₂ computed from IFTs estimated using the PC-SAFT EoS^{20,21} combined with cDFT.³⁸ The vapor spinodal limit of pure CO₂, computed using the PC-SAFT EoS.^{20,21}

T / [K]	SCL / [bar]	Liquid spinodal / [bar]
243.0	17.31	32.58
244.0	17.78	33.01
245.0	18.26	33.45
246.0	18.75	33.89
247.0	19.25	34.34
248.0	19.76	34.79
249.0	20.29	35.25
250.0	20.82	35.72
251.0	21.37	36.19
252.0	21.92	36.66
253.0	22.49	37.14
254.0	23.07	37.63
255.0	23.66	38.13
256.0	24.27	38.63
257.0	24.89	39.13
258.0	25.52	39.65
259.0	26.16	40.17
260.0	26.81	40.69
261.0	27.48	41.23
262.0	28.17	41.77
263.0	28.86	42.31
264.0	29.58	42.87
265.0	30.30	43.43

Table S41: The Supercool limit (SCL) of pure CO₂ computed from IFTs estimated using the PC-SAFT EoS^{20,21} combined with cDFT.³⁸ The vapor spinodal limit of pure CO₂, computed using the PC-SAFT EoS.^{20,21}

T / [K]	SCL / [bar]	Liquid spinodal / [bar]
266.0	31.04	44.00
267.0	31.80	44.58
268.0	32.57	45.16
269.0	33.36	45.76
270.0	34.16	46.36
271.0	34.98	46.97
272.0	35.81	47.59
272.0	35.81	48.22
273.0	36.66	48.85
274.0	37.53	49.50
275.0	38.42	50.16
276.0	39.32	50.82
277.0	40.24	51.50
278.0	41.18	52.18
279.0	42.14	52.88
280.0	43.12	53.59
281.0	44.12	54.31
282.0	45.14	55.04
283.0	46.18	55.79
284.0	47.24	56.54
285.0	48.32	57.31
286.0	49.42	58.10
287.0	50.54	58.90

Table S41: The Supercool limit (SCL) of pure CO₂ computed from IFTs estimated using the PC-SAFT EoS^{20,21} combined with cDFT.³⁸ The vapor spinodal limit of pure CO₂, computed using the PC-SAFT EoS.^{20,21}

T / [K]	SCL / [bar]	Liquid spinodal / [bar]
288.0	51.69	59.71
289.0	52.86	60.54
290.0	54.06	61.38
291.0	55.28	62.24
292.0	56.52	63.12
293.0	57.79	64.01
294.0	59.08	64.93
295.0	60.41	65.86
296.0	61.75	66.82
297.0	63.13	67.80
298.0	64.54	68.80
299.0	65.97	69.83
300.0	67.44	70.88
301.0	68.93	71.97
302.0	70.46	73.09
303.0	72.02	74.25
304.0	73.61	75.44
305.0	75.24	76.68
306.0	76.91	77.98
307.0	78.60	79.34
309.0	82.11	82.38

Table S42: The Superheat limit (SHL) and SCL of pure CO₂ computed from IFTs estimated using the Guggenheim–Katayama equation¹³ combined with Span and Wagner EoS.¹⁹ Values of the SHL less than zero for the corresponding temperatures are indicated as ‘-’.

T / [K]	SHL / [bar]	SCL / [bar]
220.0	-	8.98
221.0	-	9.27
222.0	-	9.56
223.0	-	9.85
224.0	-	10.16
225.0	-	10.47
226.0	-	10.78
227.0	-	11.11
228.0	-	11.44
229.0	-	11.78
230.0	-	12.13
231.0	-	12.49
232.0	-	12.85
233.0	-	13.22
234.0	-	13.60
235.0	-	13.99
236.0	-	14.39
237.0	-	14.80
238.0	-	15.21
239.0	-	15.64
240.0	-	16.07
241.0	-	16.51
242.0	-	16.97

Table S42: The Superheat limit (SHL) and SCL of pure CO₂ computed from IFTs estimated using the Guggenheim–Katayama equation¹³ combined with Span and Wagner EoS.¹⁹ Values of the SHL less than zero for the corresponding temperatures are indicated as ‘-’.

T / [K]	SHL / [bar]	SCL / [bar]
243.0	-	17.43
244.0	-	17.90
245.0	-	18.38
246.0	-	18.87
247.0	-	19.37
248.0	-	19.89
249.0	-	20.41
250.0	-	20.94
251.0	-	21.49
252.0	-	22.04
253.0	-	22.61
254.0	-	23.19
255.0	-	23.78
256.0	-	24.38
257.0	-	25.00
258.0	-	25.62
259.0	-	26.26
260.0	-	26.92
261.0	-	27.58
262.0	-	28.26
263.0	-	28.95
264.0	-	29.66
265.0	-	30.38

Table S42: The Superheat limit (SHL) and SCL of pure CO₂ computed from IFTs estimated using the Guggenheim–Katayama equation¹³ combined with Span and Wagner EoS.¹⁹ Values of the SHL less than zero for the corresponding temperatures are indicated as ‘-’.

T / [K]	SHL / [bar]	SCL / [bar]
266.0	-	31.11
267.0	-	31.86
268.0	-	32.63
269.0	-	33.41
270.0	1.65	34.20
271.0	4.23	35.01
272.0	6.77	35.84
273.0	9.28	36.68
274.0	11.77	37.54
275.0	14.22	38.42
276.0	16.64	39.31
277.0	19.03	40.22
278.0	21.39	41.15
279.0	23.73	42.10
280.0	26.03	43.06
281.0	28.31	44.05
282.0	30.55	45.06
283.0	32.78	46.08
284.0	34.97	47.13
285.0	37.13	48.20
286.0	39.27	49.29
287.0	41.38	50.40
288.0	43.47	51.53

Table S42: The Superheat limit (SHL) and SCL of pure CO₂ computed from IFTs estimated using the Guggenheim–Katayama equation¹³ combined with Span and Wagner EoS.¹⁹ Values of the SHL less than zero for the corresponding temperatures are indicated as ‘-’.

T / [K]	SHL / [bar]	SCL / [bar]
289.0	45.53	52.69
290.0	47.57	53.88
291.0	49.58	55.08
292.0	51.56	56.32
293.0	53.52	57.58
294.0	55.46	58.86
295.0	57.37	60.18
296.0	59.26	61.53
297.0	61.12	62.90
298.0	62.96	64.31
299.0	64.78	65.75
300.0	66.58	67.23
301.0	68.36	68.74
302.0	70.11	70.30
303.0	71.84	71.90
304.0	73.55	73.56

Table S43: The Superheat limit (SHL) and SCL of pure CO₂ computed from IFTs estimated from MD simulations. Values of the SHL less than zero for the corresponding temperatures are indicated as ‘-’.

T /[K]	SHL / [bar]	SCL / [bar]
220.0	-	9.08
230.0	-	12.22
240.0	-	16.10

Table S43: The Superheat limit (SHL) and SCL of pure CO₂ computed from IFTs estimated from MD simulations. Values of the SHL less than zero for the corresponding temperatures are indicated as ‘-’.

$T/[K]$	SHL / [bar]	SCL / [bar]
250.0	-	20.86
260.0	-	26.87
270.0	3.26	34.08
280.0	27.61	42.91
290.0	49.41	53.65

Table S44: List of .csv files containing raw SHL, SCL, and saturation lines (dew and bubble lines) data of binary mixtures containing 95% CO₂ and 5% impurity (CH₄, Ar, N₂, or H₂). The SHL and SCL lines of CO₂-CH₄, CO₂-Ar, and CO₂-N₂ are computed using IFTs obtained from the Parachor model.⁴² For CO₂-H₂, the SHL and SCL lines are computed using IFTs obtained from the WSD model.⁴³ The saturation lines were computed using the GERG-2008 EoS.⁴⁴

Binary mixture	Superheat limit (SHL)	Supercool limit (SCL)	Saturation lines
CO ₂ -CH ₄	SHL_CO2_CH4.csv	SCL_CO2_CH4.csv	CO2_CH4-GERG2008.csv
CO ₂ -Ar	SHL_CO2_Ar.csv	SCL_CO2_Ar.csv	CO2_Ar-GERG2008.csv
CO ₂ -N ₂	SHL_CO2_N2.csv	SCL_CO2_N2.csv	CO2_N2-GERG2008.csv
CO ₂ -H ₂	SHL_CO2_H2.csv	SCL_CO2_H2.csv	CO2_H2-GERG2008.csv

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